

Introduction to Conservation Genomics

Mpala workshop, August 2026

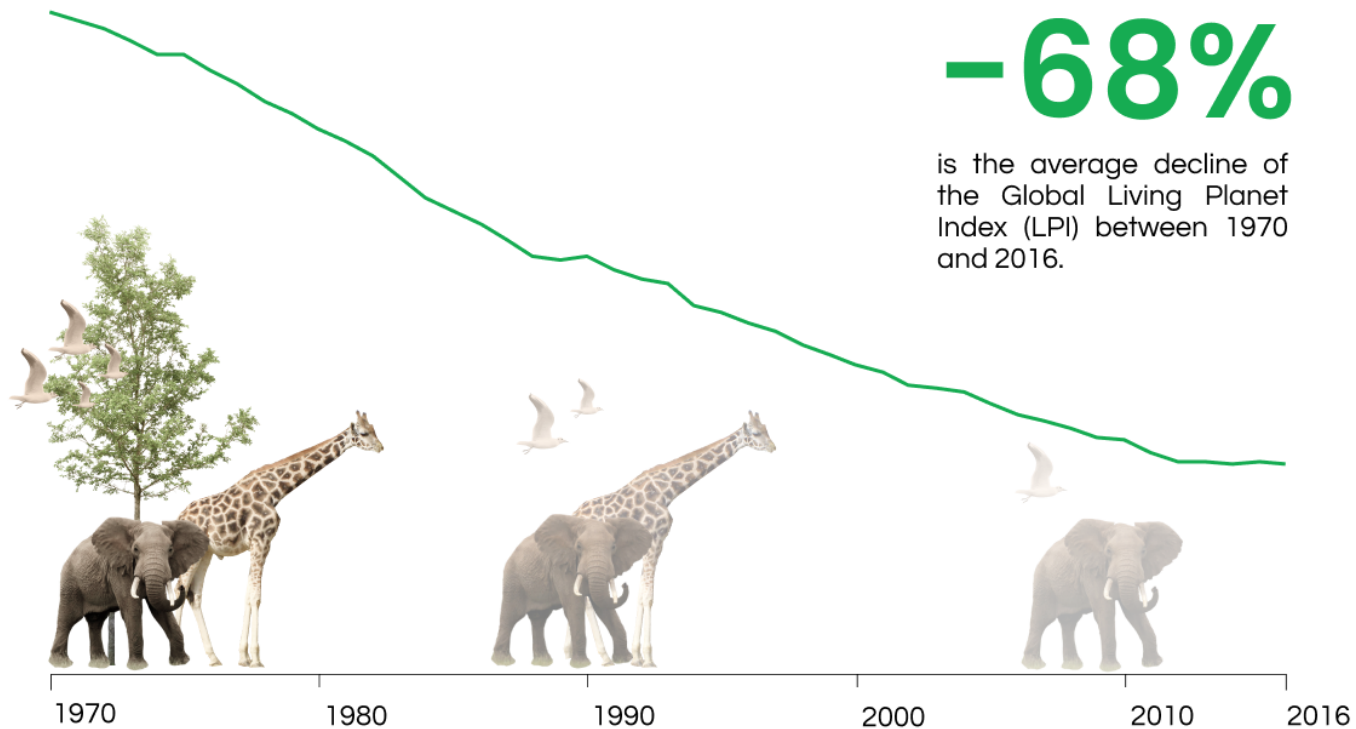


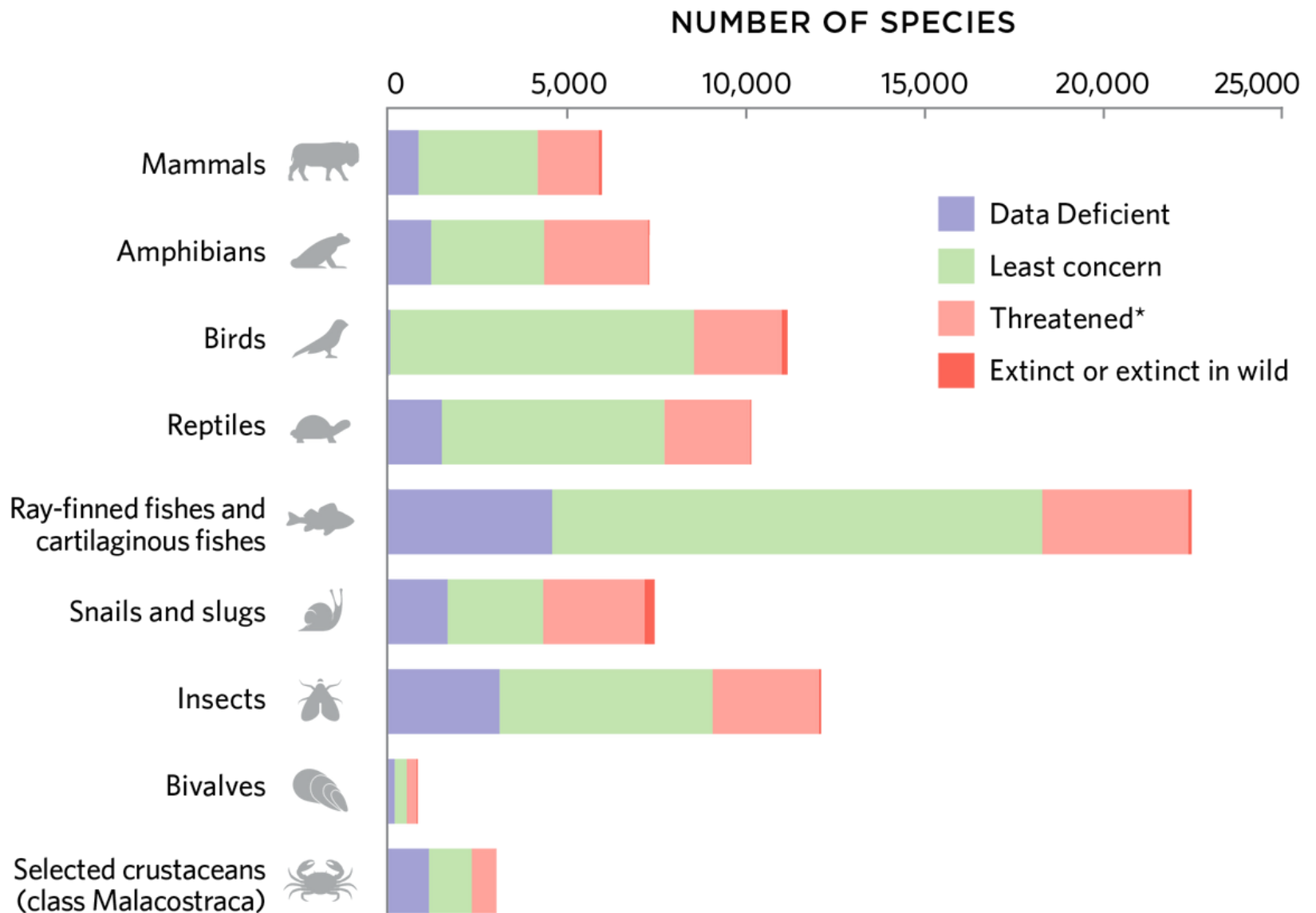


Wild species are in decline everywhere

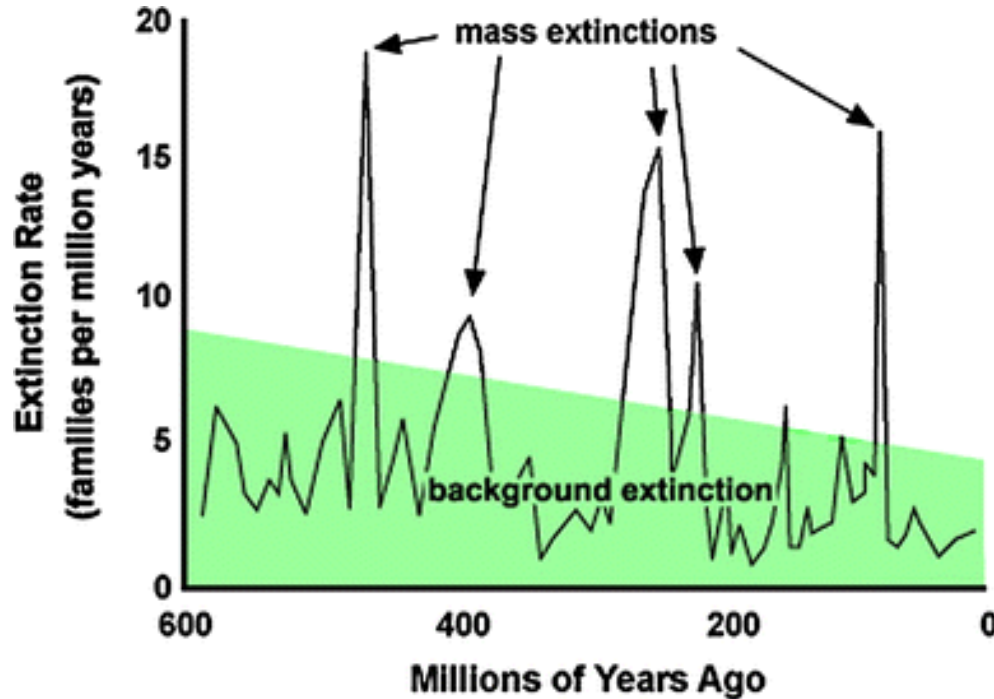
Figure 1. Living Planet Index, World

Source: World Wildlife Fund (WWF) and Zoological Society of London.



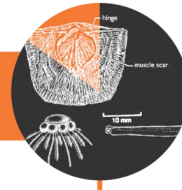


Source: IUCN Red List



LATE ORDOVICIAN

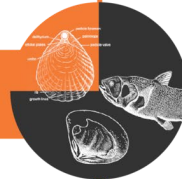
440 MILLION
YEARS AGO



85%
EXTINCT

LATE DEVONIAN

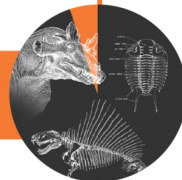
374 MILLION
YEARS AGO



75%
EXTINCT

PERMIAN

250 MILLION
YEARS AGO



95%
EXTINCT

JURASSIC

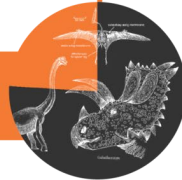
200 MILLION
YEARS AGO



80%
EXTINCT

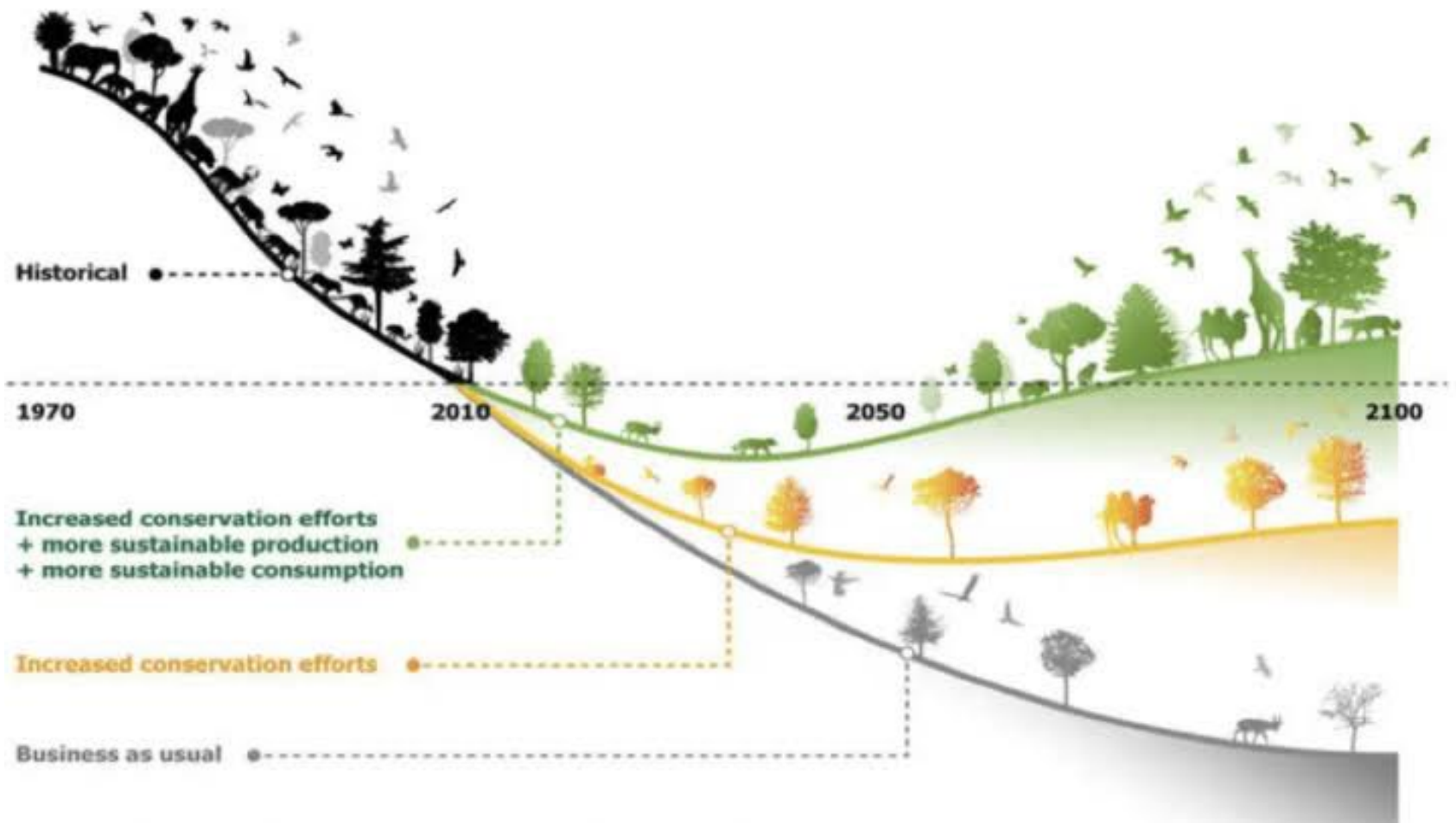
CRETACEOUS

65 MILLION
YEARS AGO



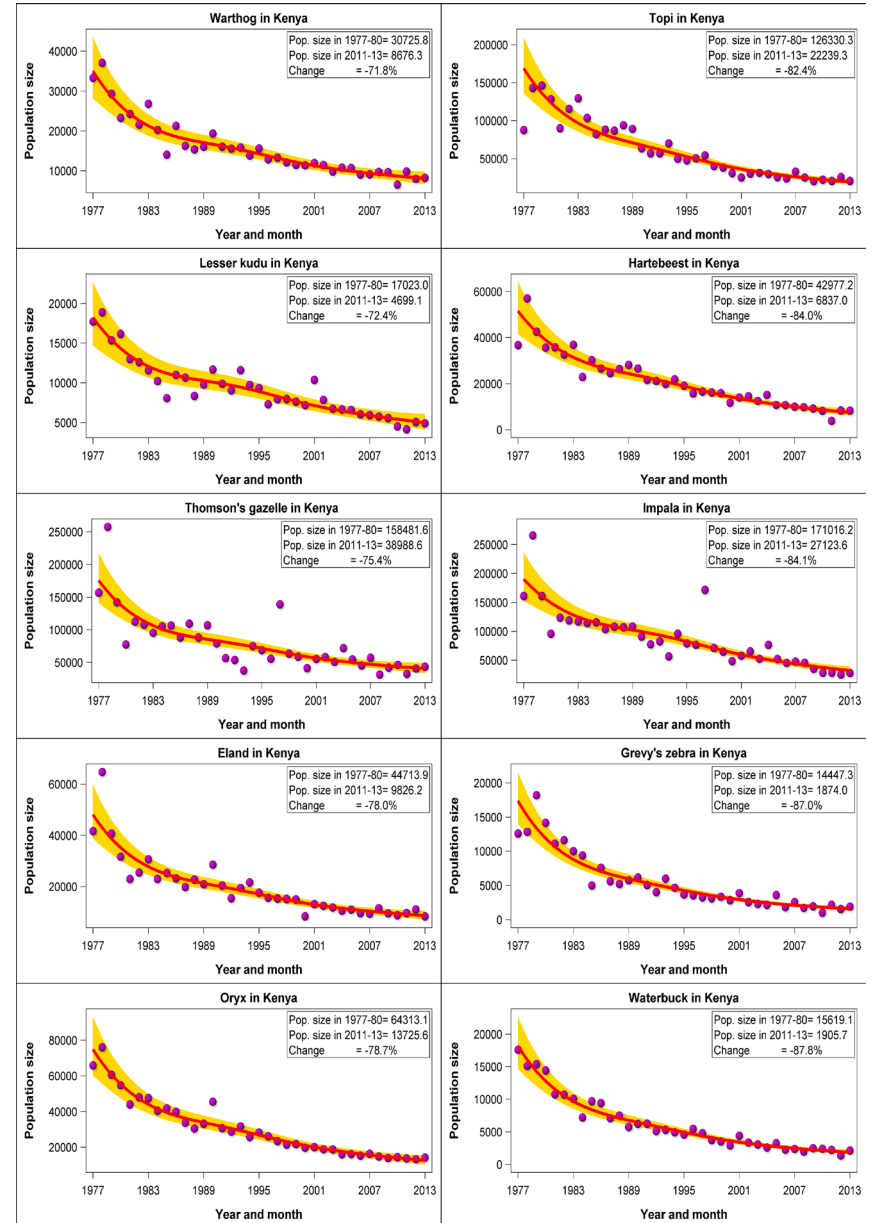
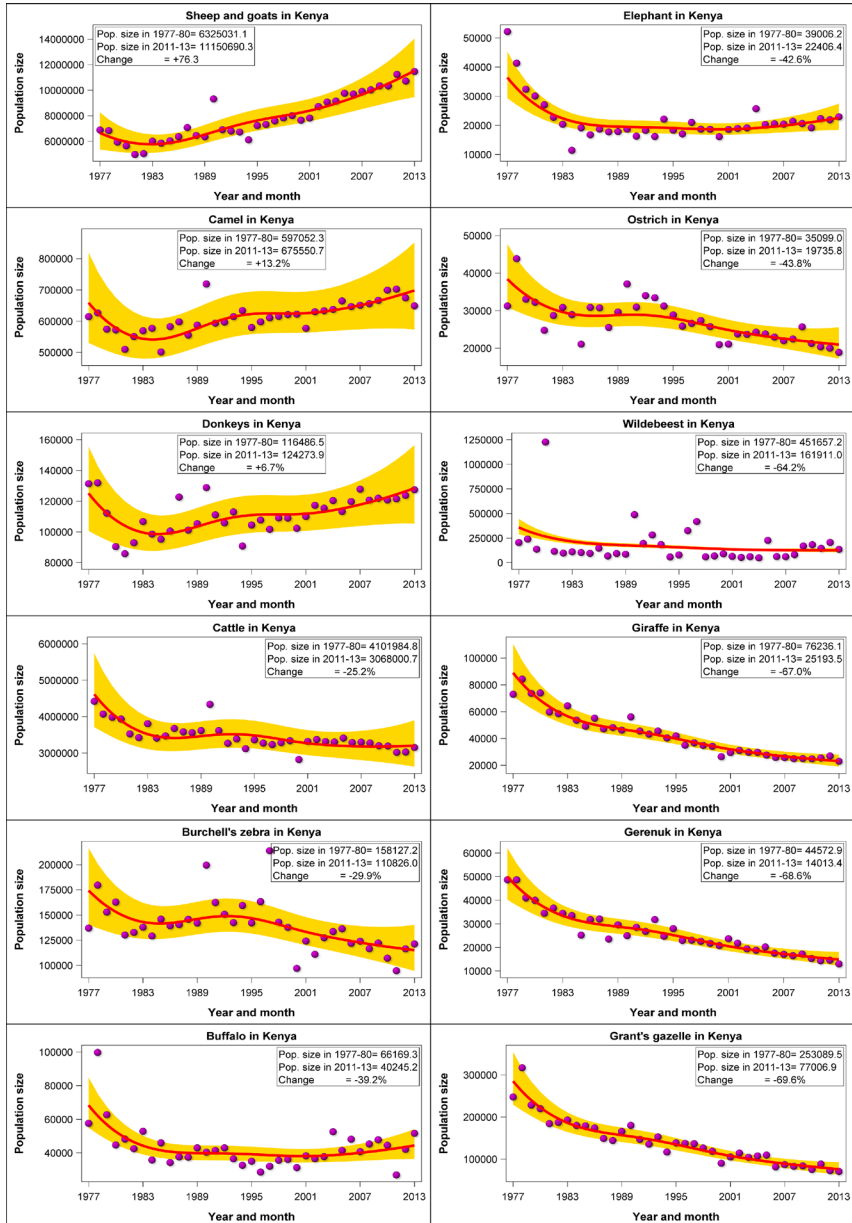
76%
EXTINCT

Are we in the 6th Mass Extinction?

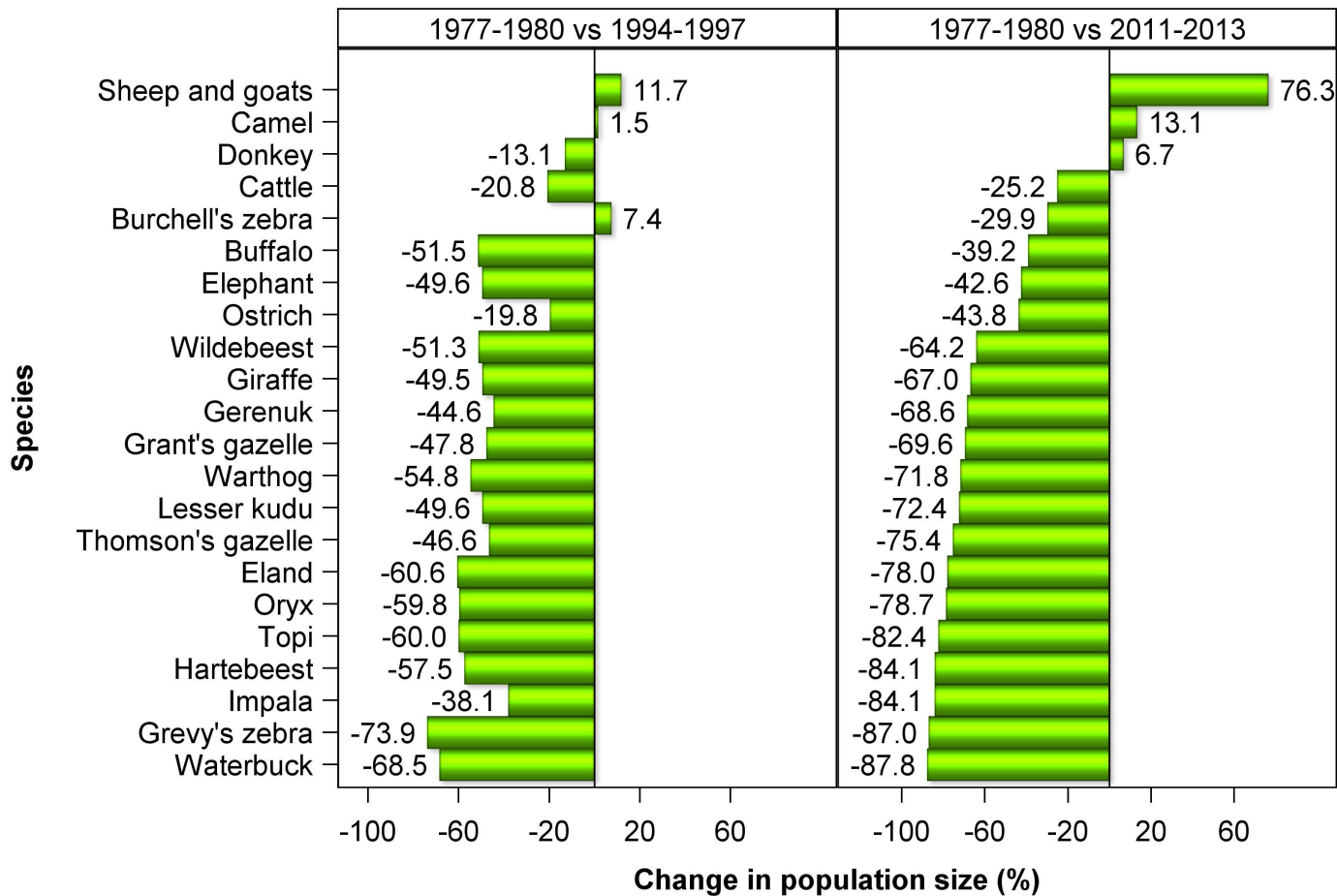


We can still reverse the decline

What about Kenya?



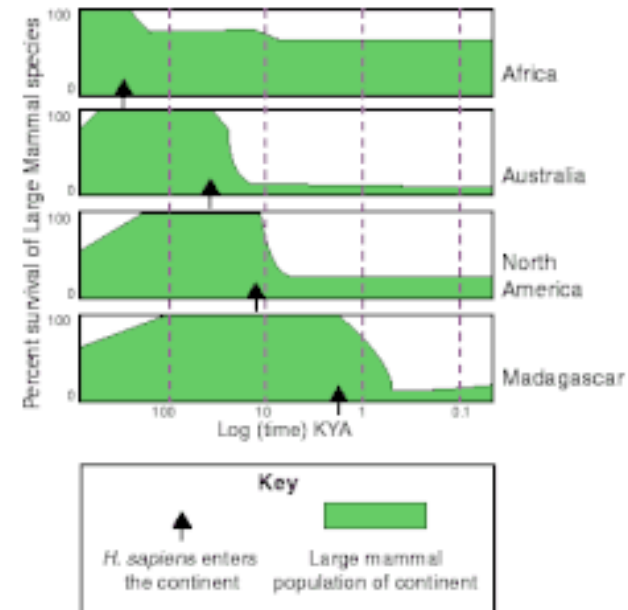
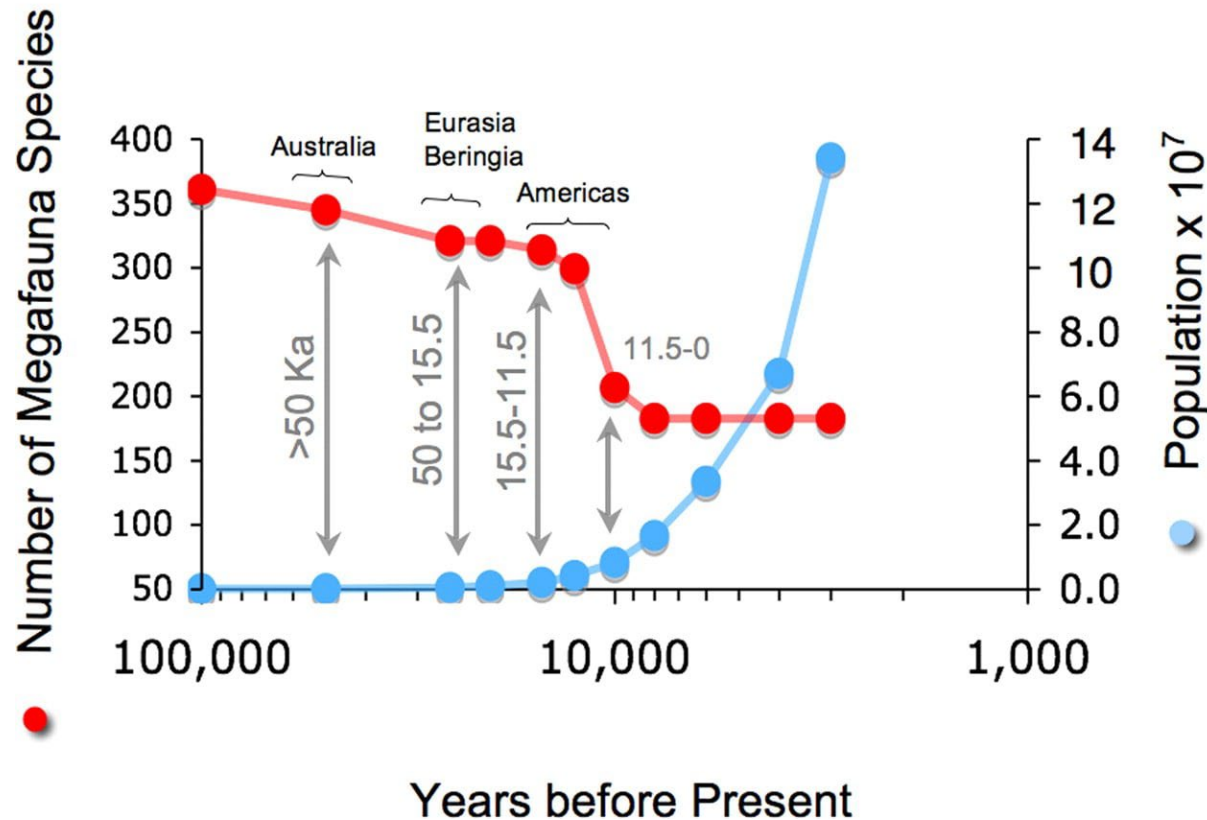
Change in population size (%)



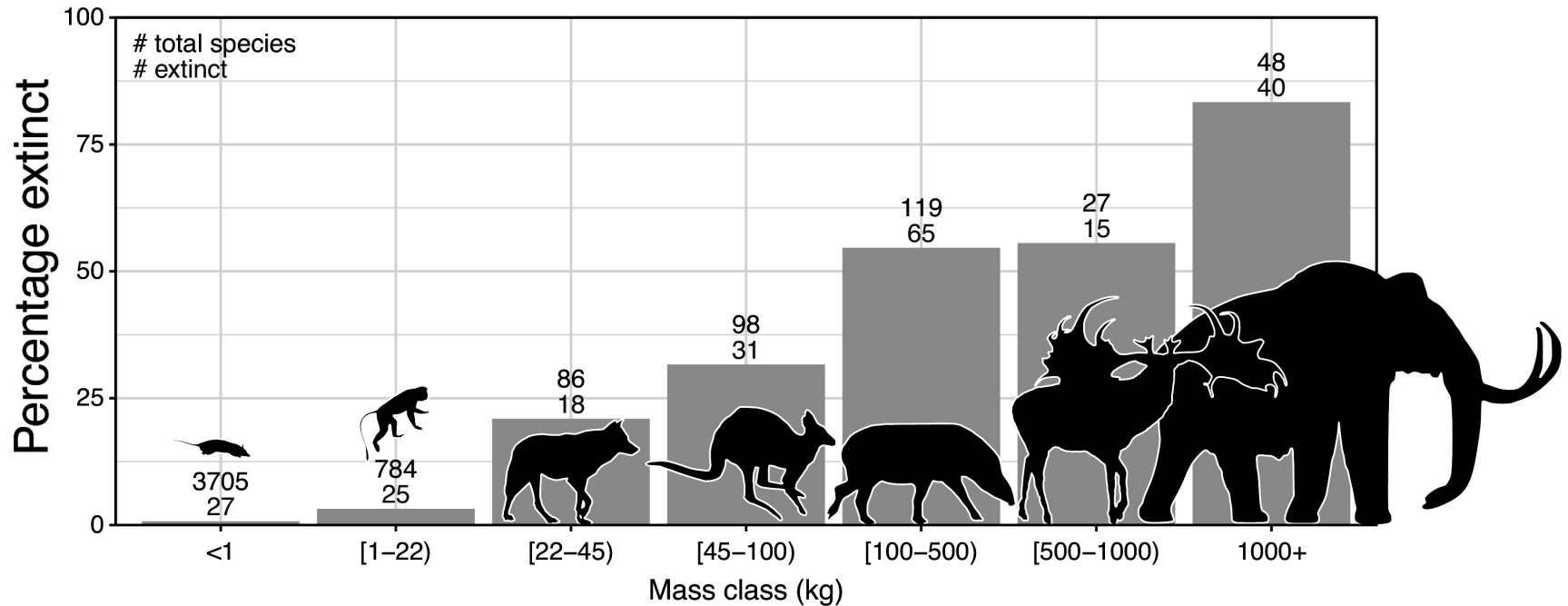
Ogutu et al. 2016

When did we start losing wild animal species?

Megafauna Loss vs. Global Human Population Growth



Being a large animal is dangerous

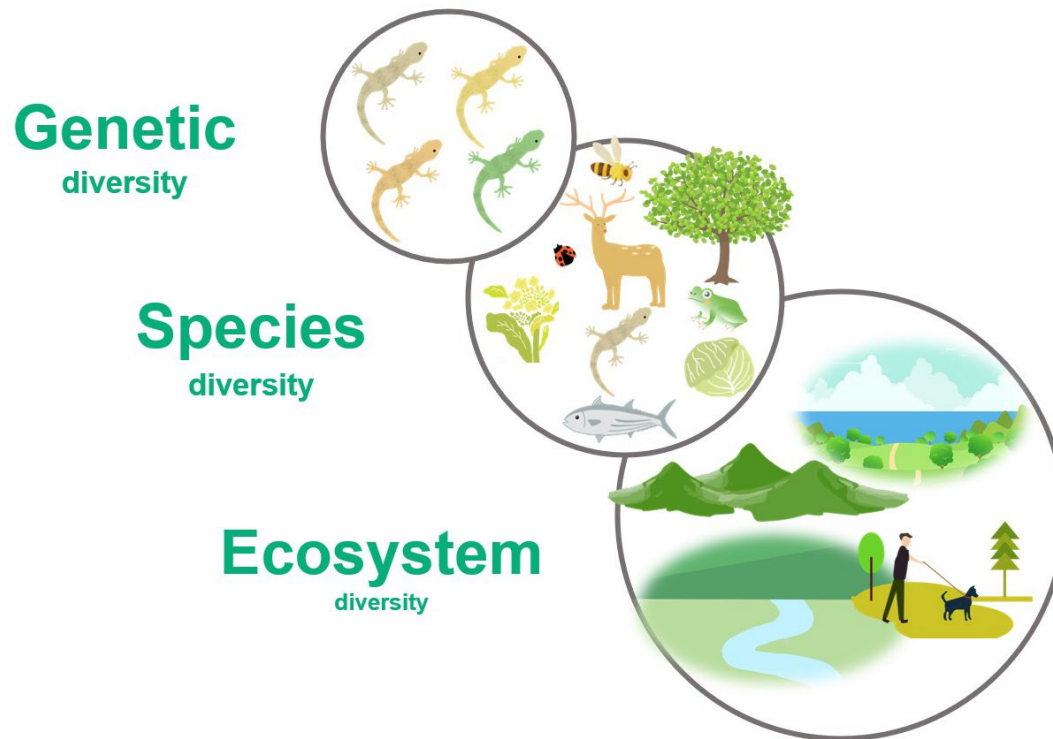


Species extinction within the last 50,000 years

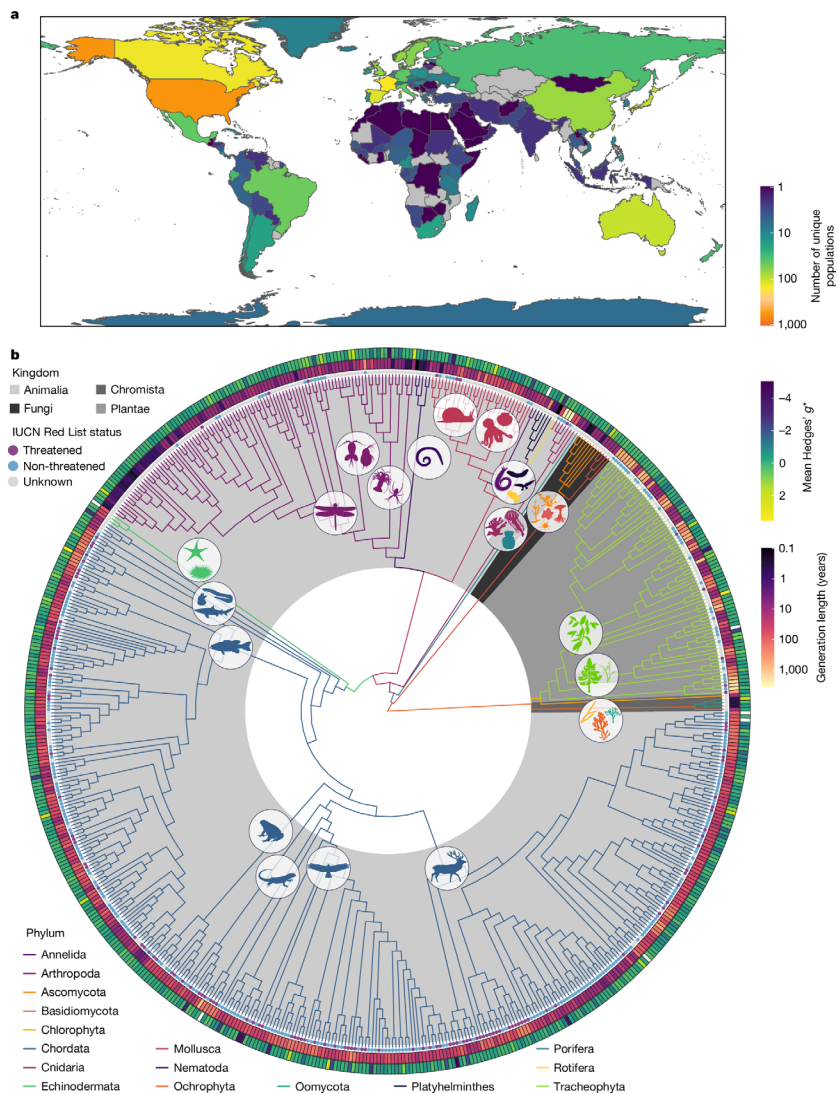
Svenning et al. 2024

Biodiversity is not just species richness

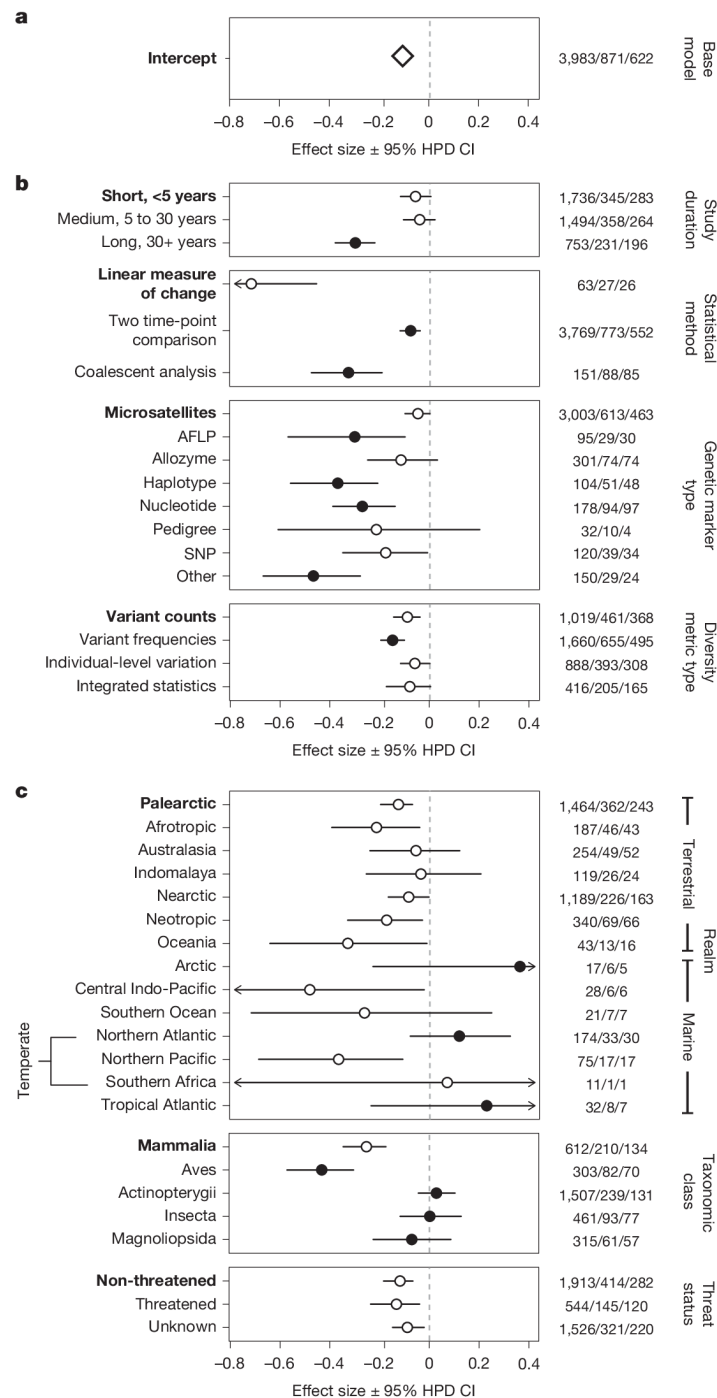
Biodiversity



The UN Convention on Biological Diversity recognises genetic diversity as a key biodiversity component that **must be protected**

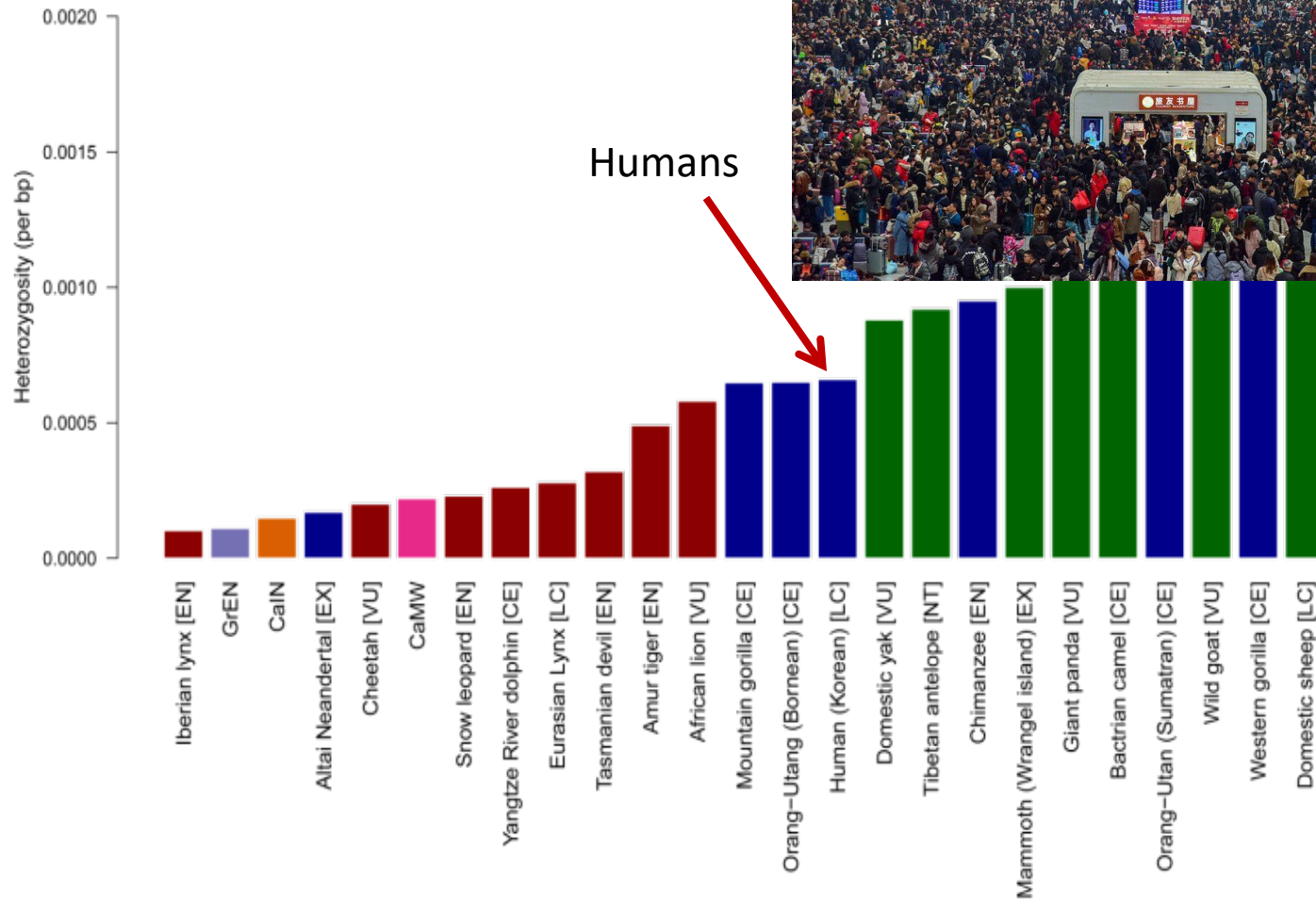


Shaw et al. 2025



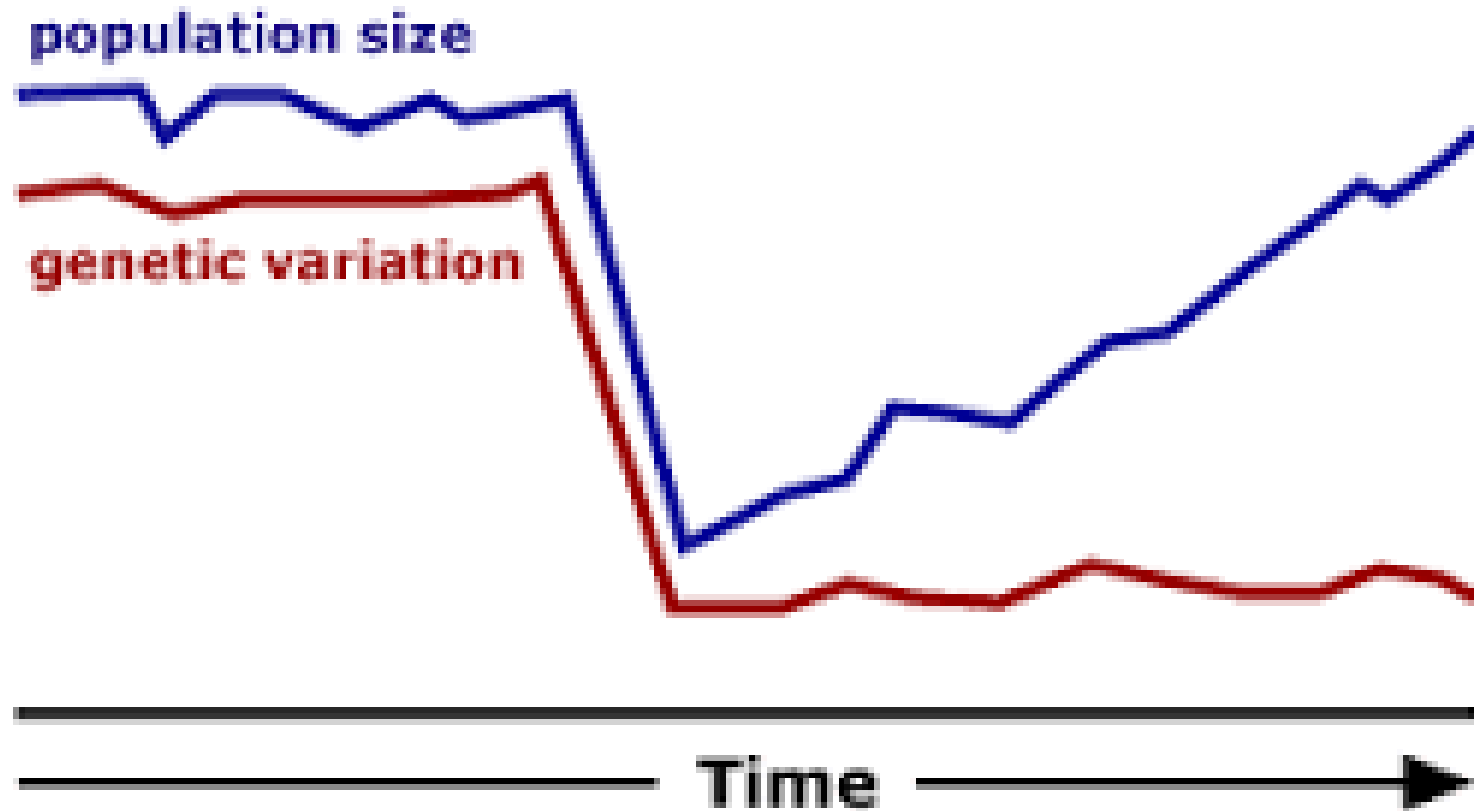
How does
genetic diversity
relate to
population sizes
over time?





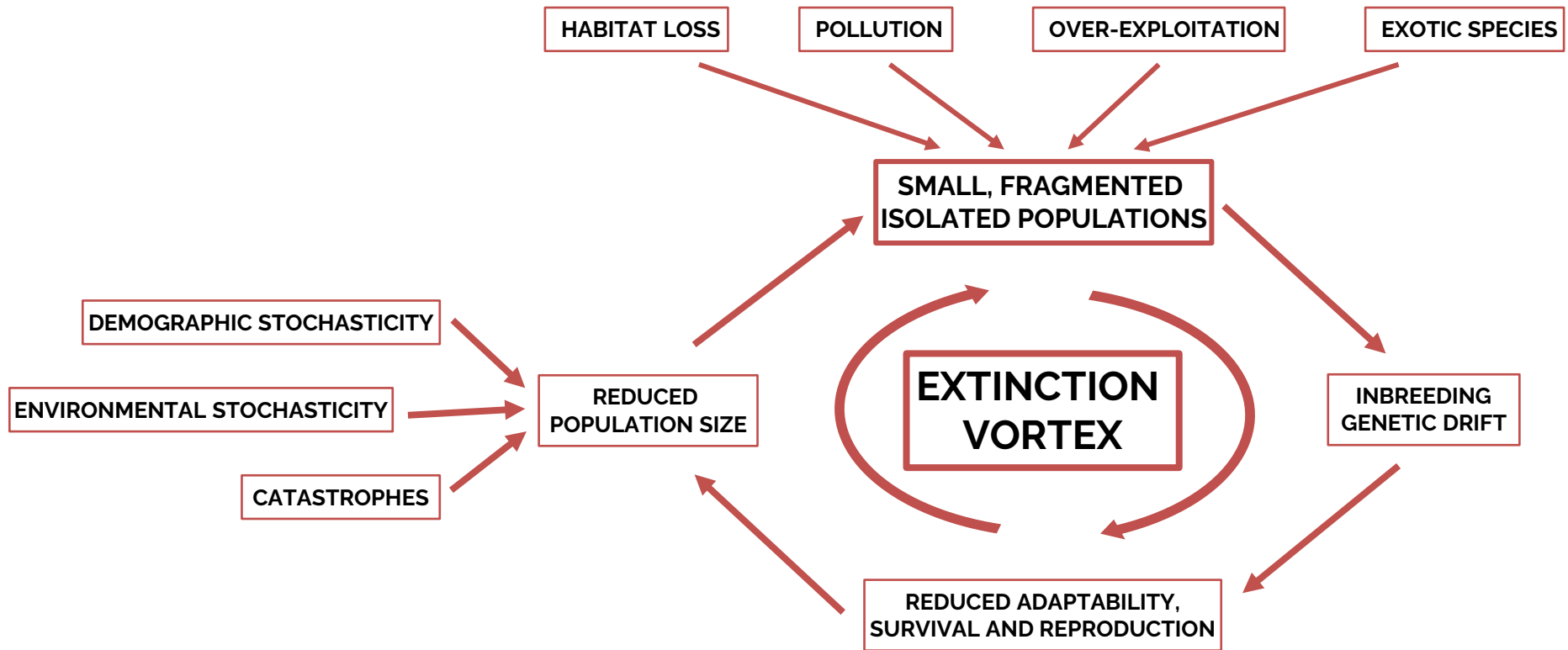
Why do humans have low genetic diversity?

(3.8)
$$N_e = \frac{1}{p_1/N + p_2/N_2 + \dots p_k/N_k}$$

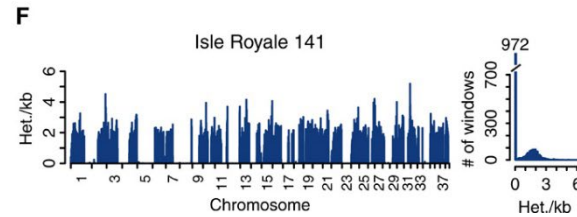
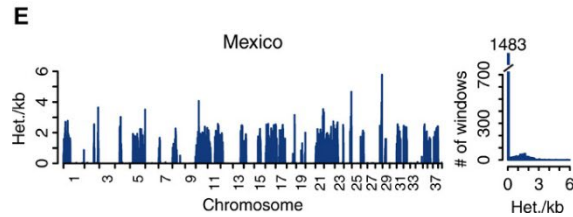
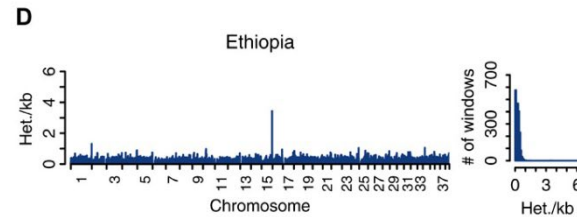
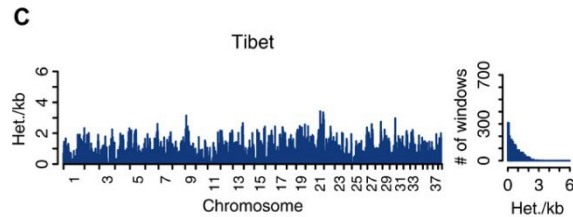
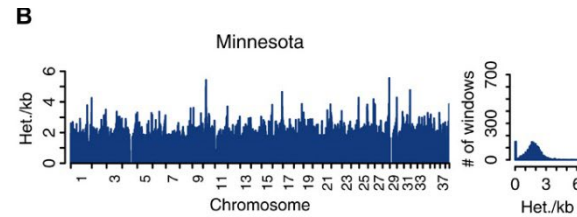
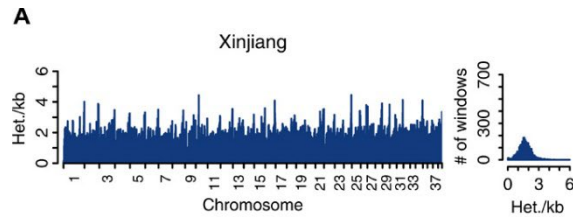
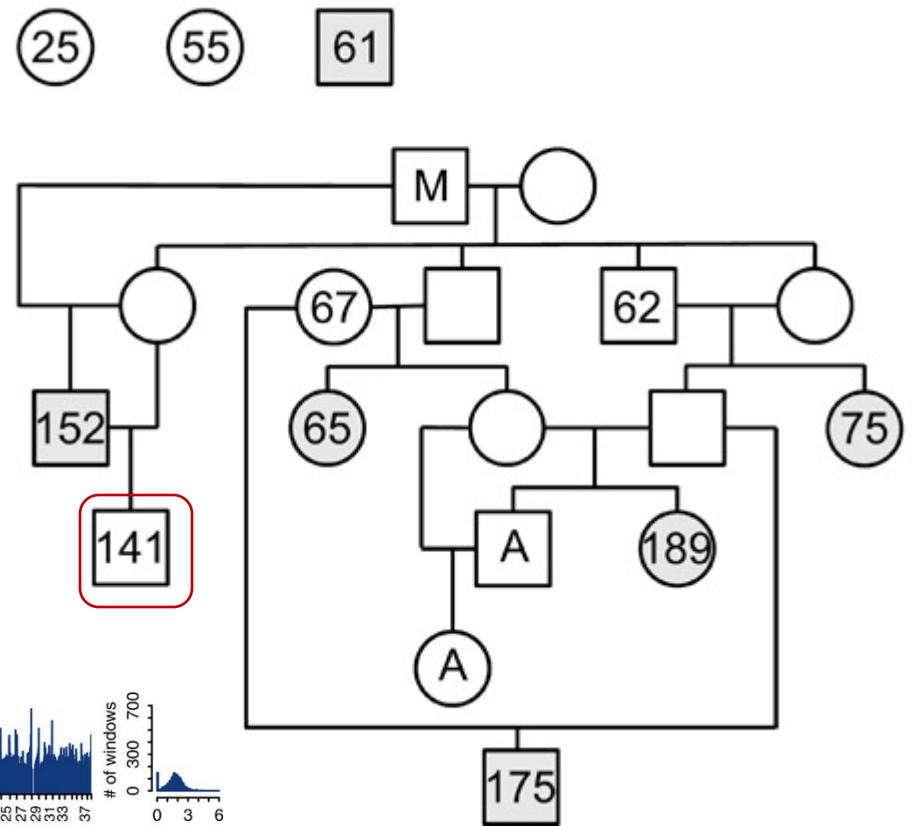


Small population sizes matter more than large ones

The predicament of low diversity

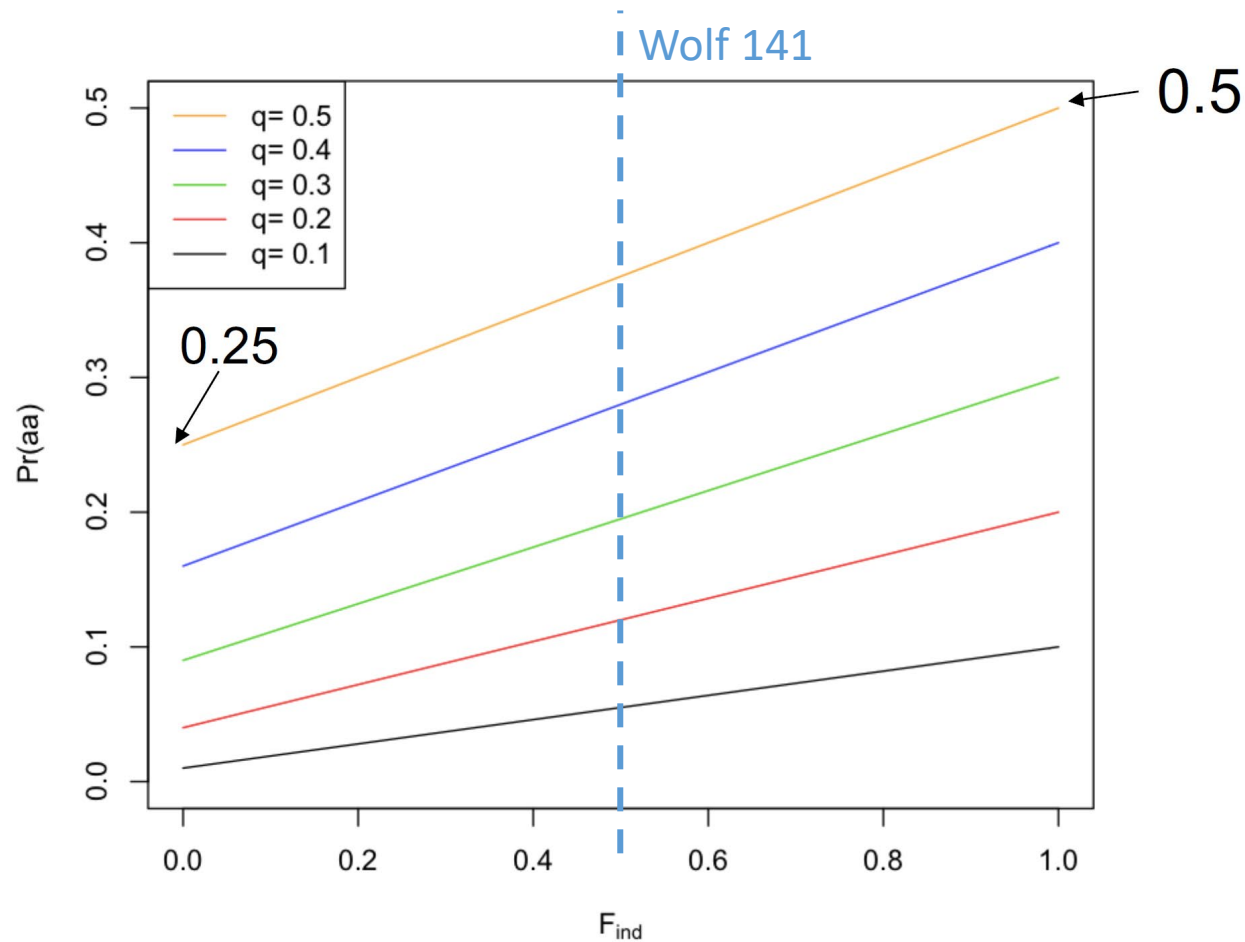


Small populations easily get in trouble



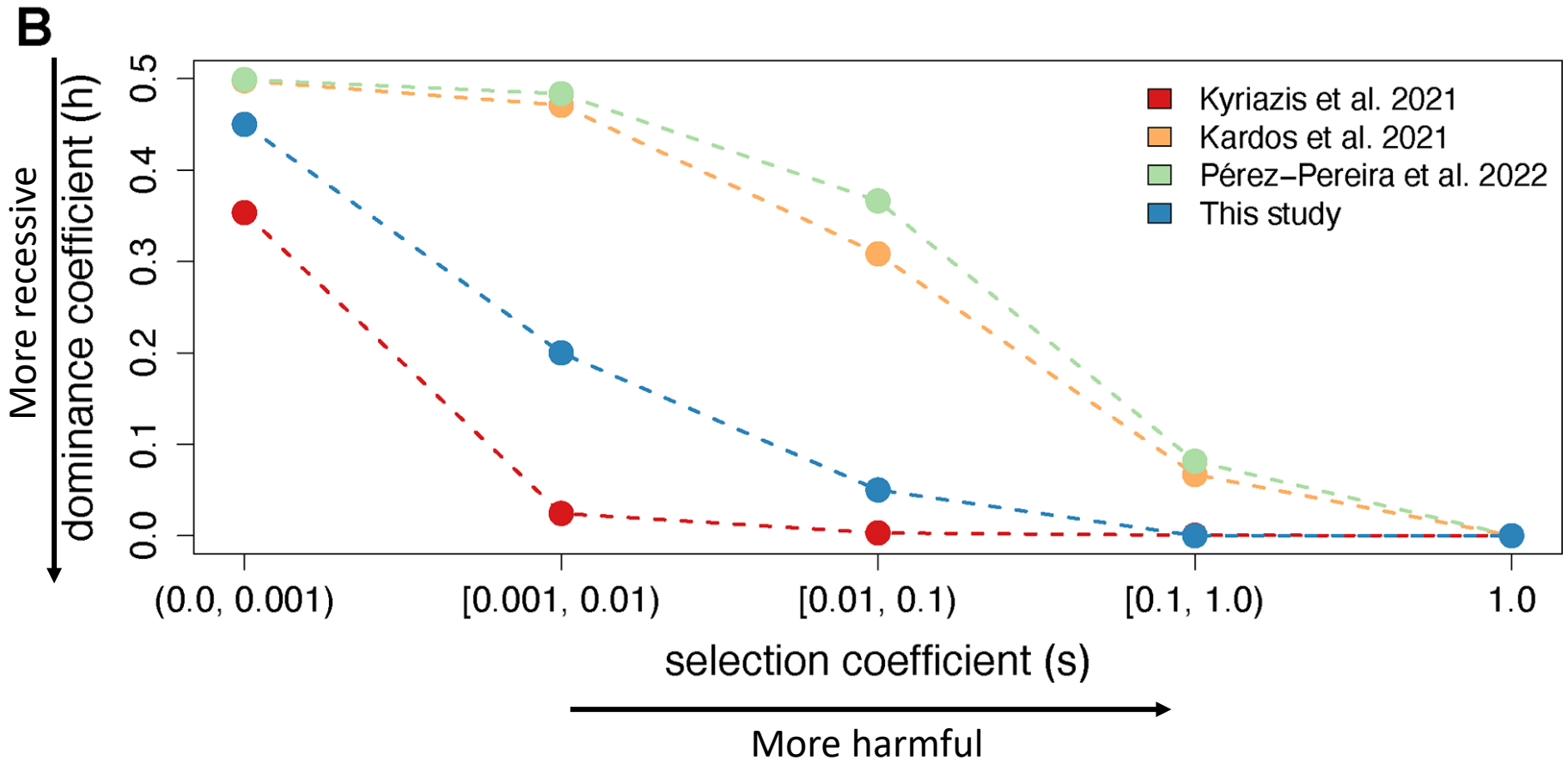
Wolf 141 is inbred in almost half of the genome

Inbreeding exposes recessive alleles



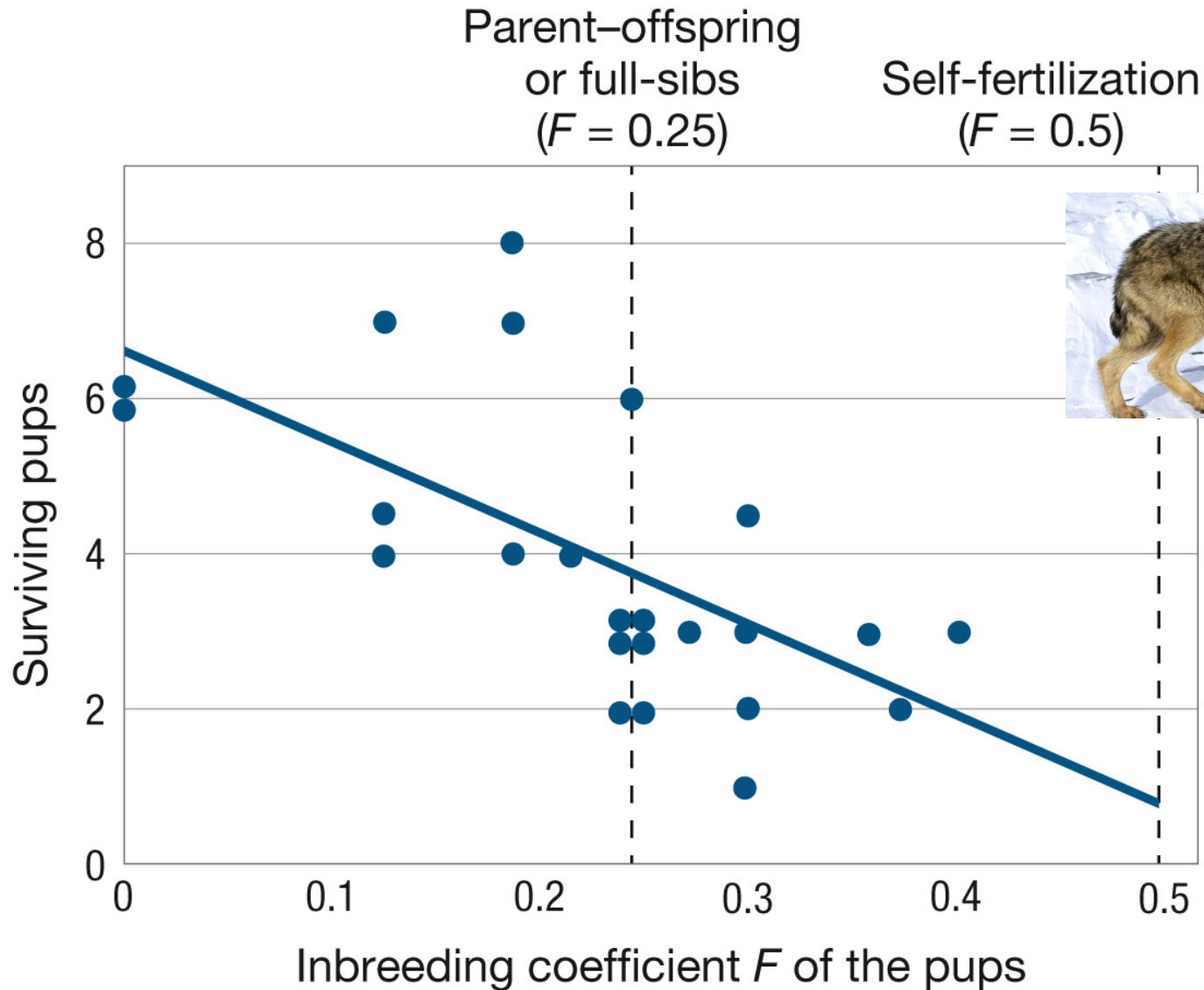
Probability of being aa:

$$(1 - F_{ind})q^2 + F_{ind}q$$



Very harmful alleles tend to be recessive, and very recessive alleles tend to be harmful.

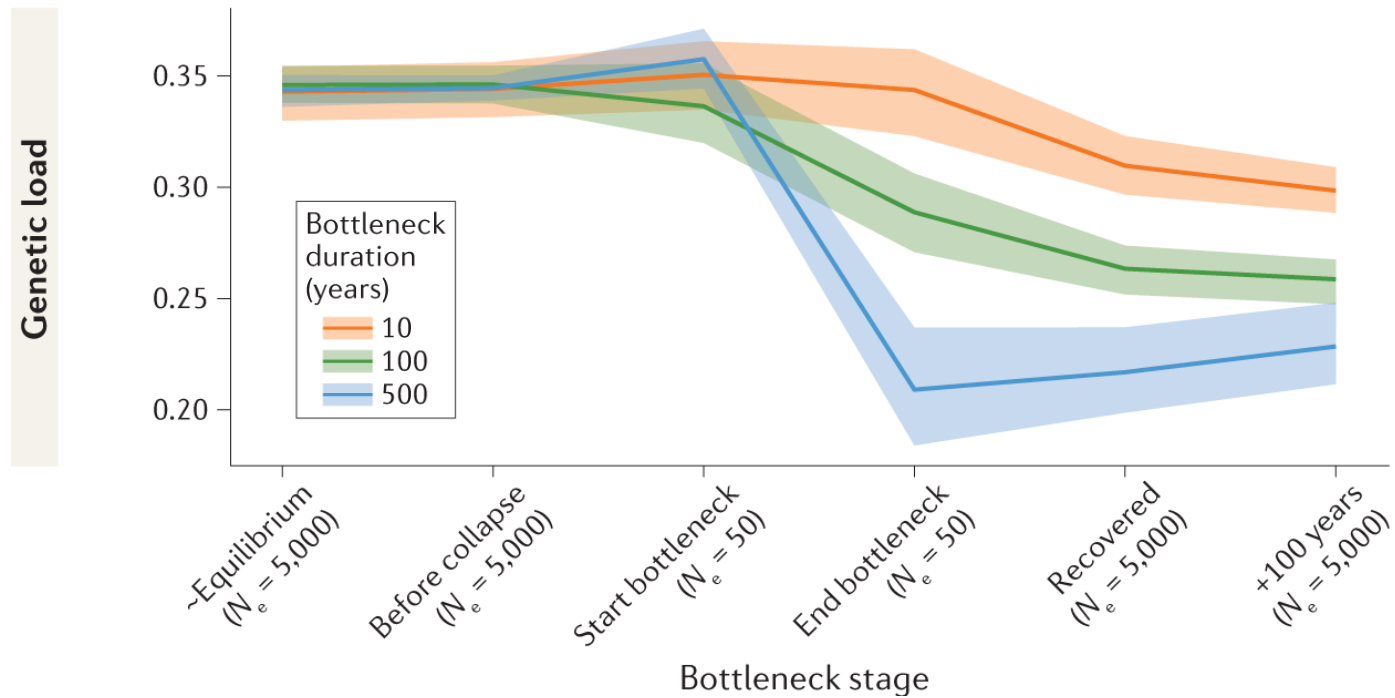
Inbreeding depression is real!



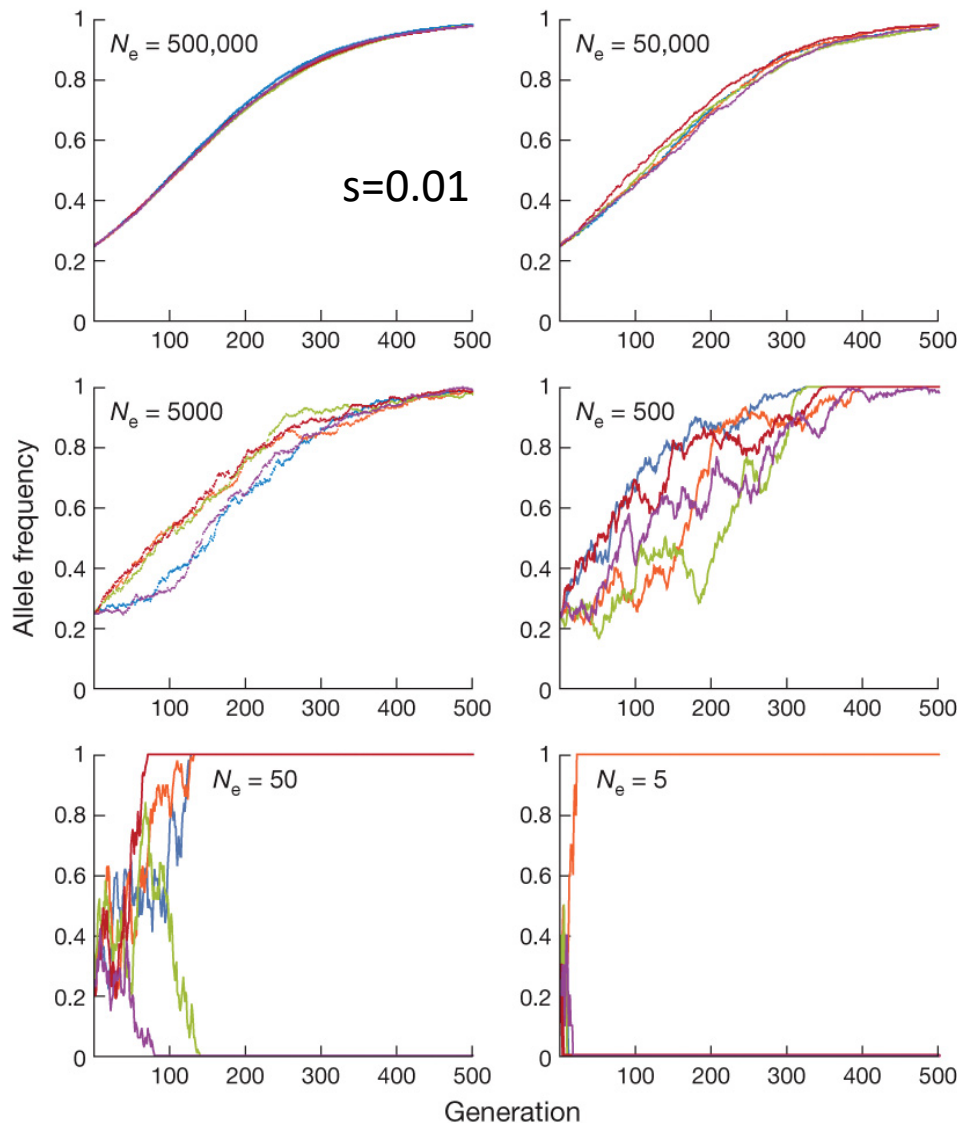
Genetic Load

The relative amount by which the average fitness of a POPULATION is lowered, due to the presence of GENES that decrease survival, compared to the GENOTYPE with maximum or optimal fitness. (From Rieger et al., Glossary of Genetics: Classical and Molecular, 5th ed)

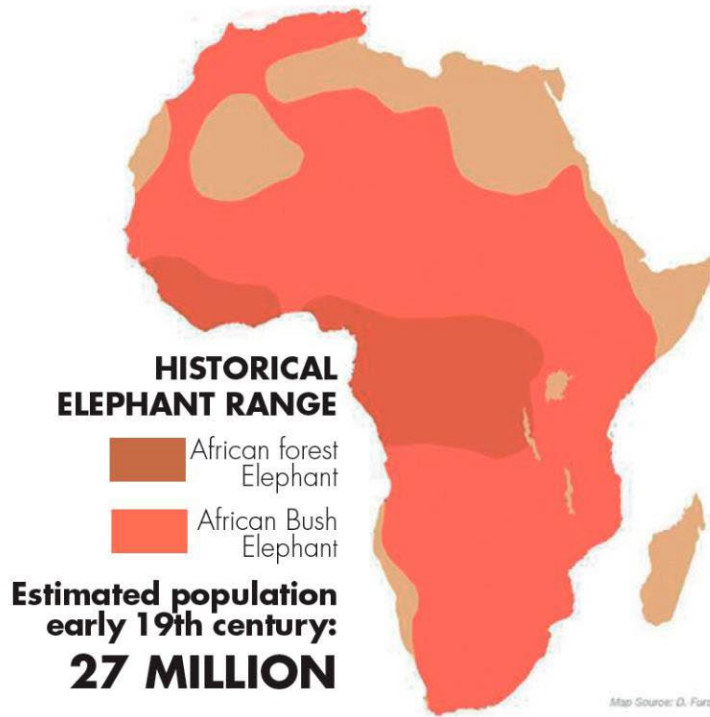
Genetic load depends on history



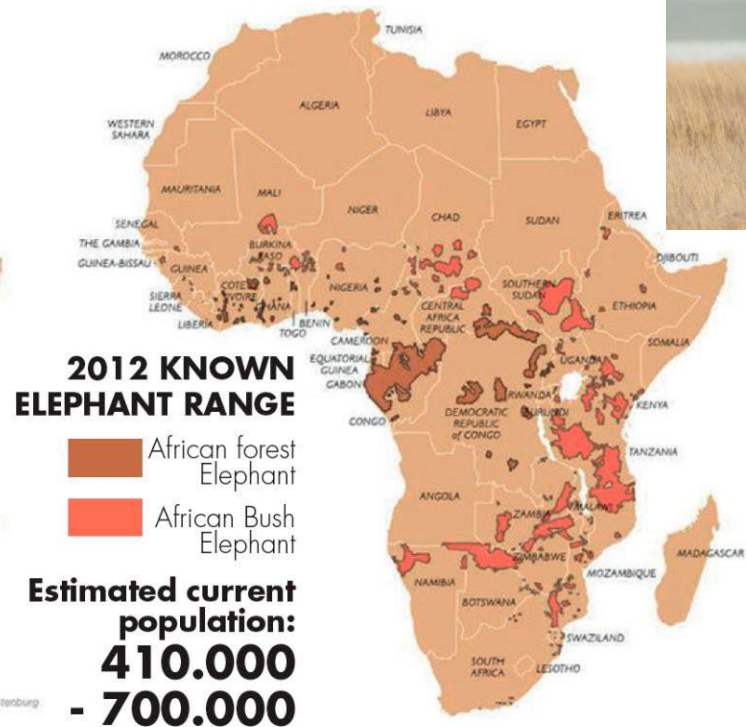
Selection less efficient in small populations



- In large populations, selection always “wins”.
- In small populations, drift usually wins.
- Drift makes fixation or loss occur faster.
- Even beneficial mutations often get lost!

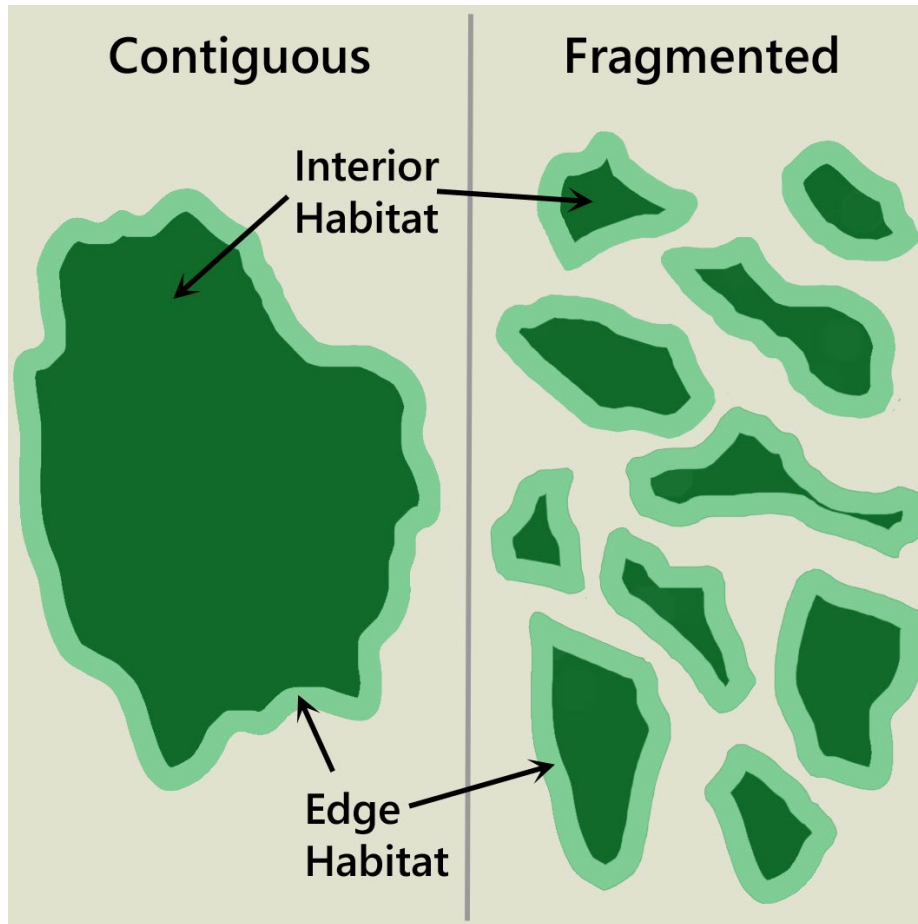


Source: Elephant Action League EDGE of Existence

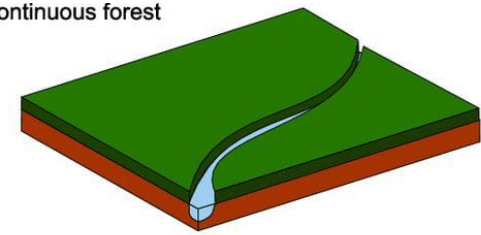


Source: IUCN African Elephant Specialist Group



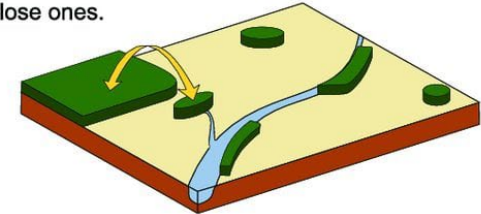


Very large continuous forest



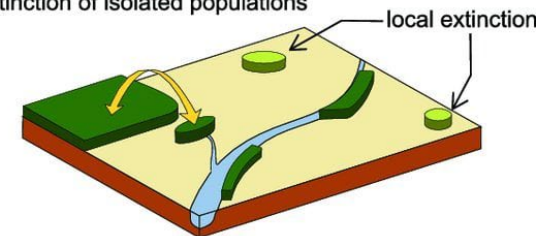
Fragmentation

No gene flow between the large conserved area and distant remnants, and low gene flow to the close ones.



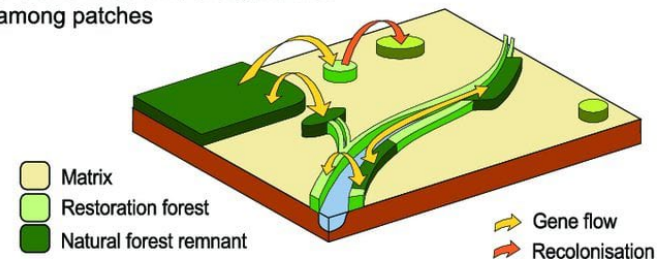
Time

Strong genetic drift effects on small populations and local extinction of isolated populations

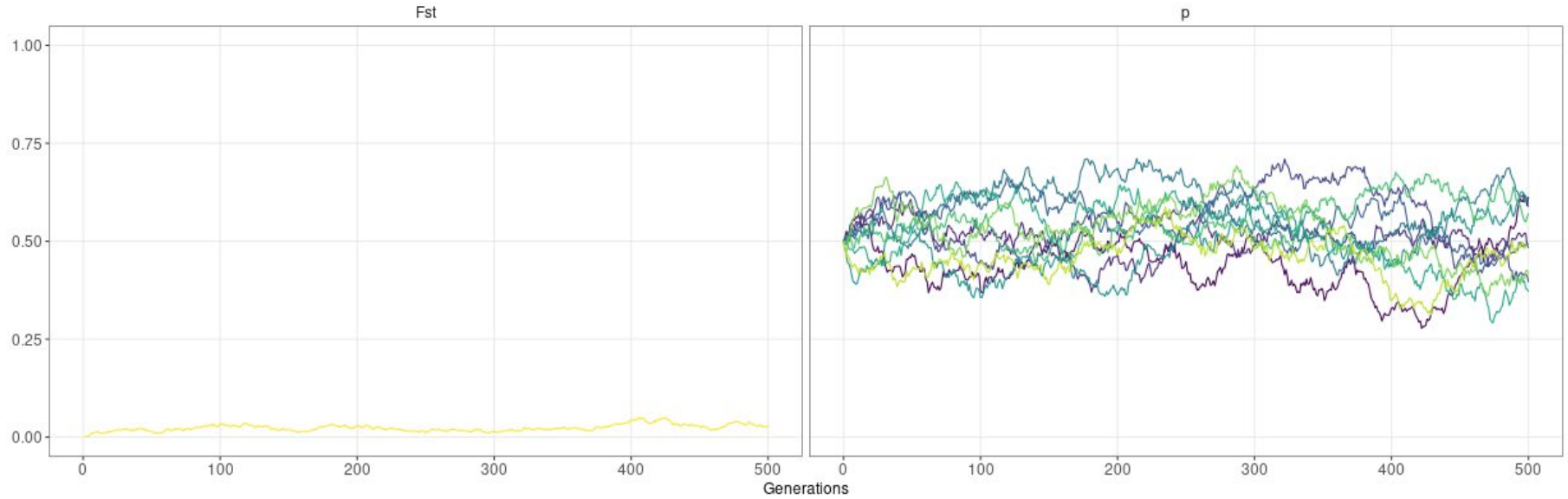


Restoration

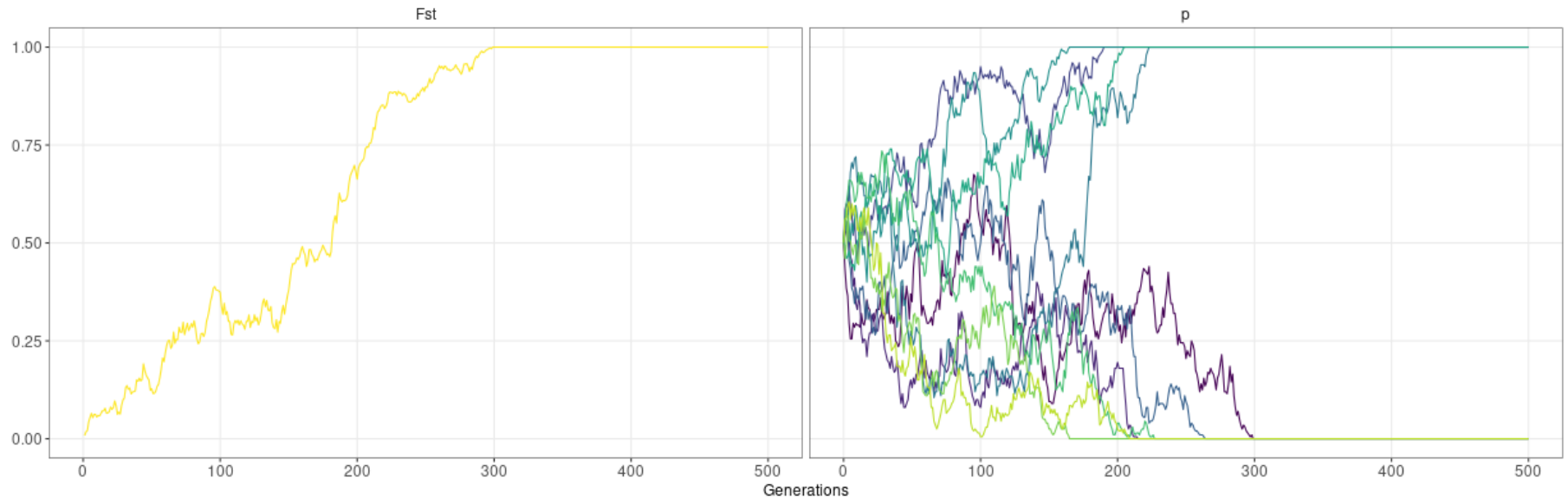
Increase in total genetic diversity, recolonisation of some fragments and gene flow among patches



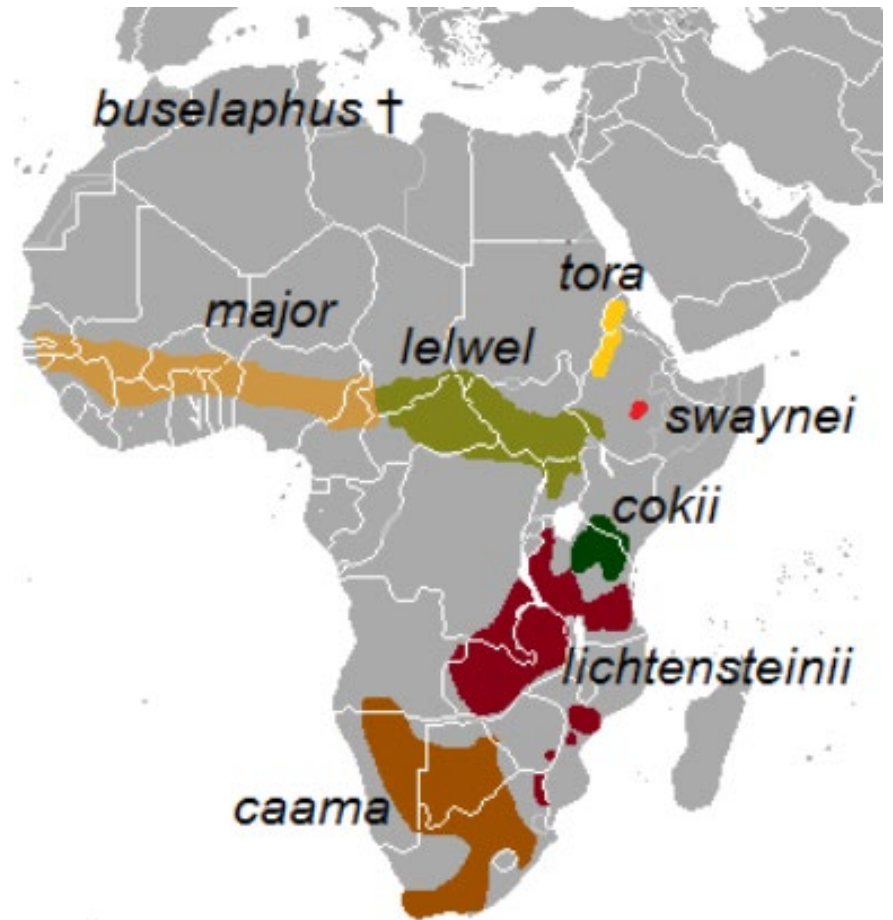
$N=1000, m=0.01$

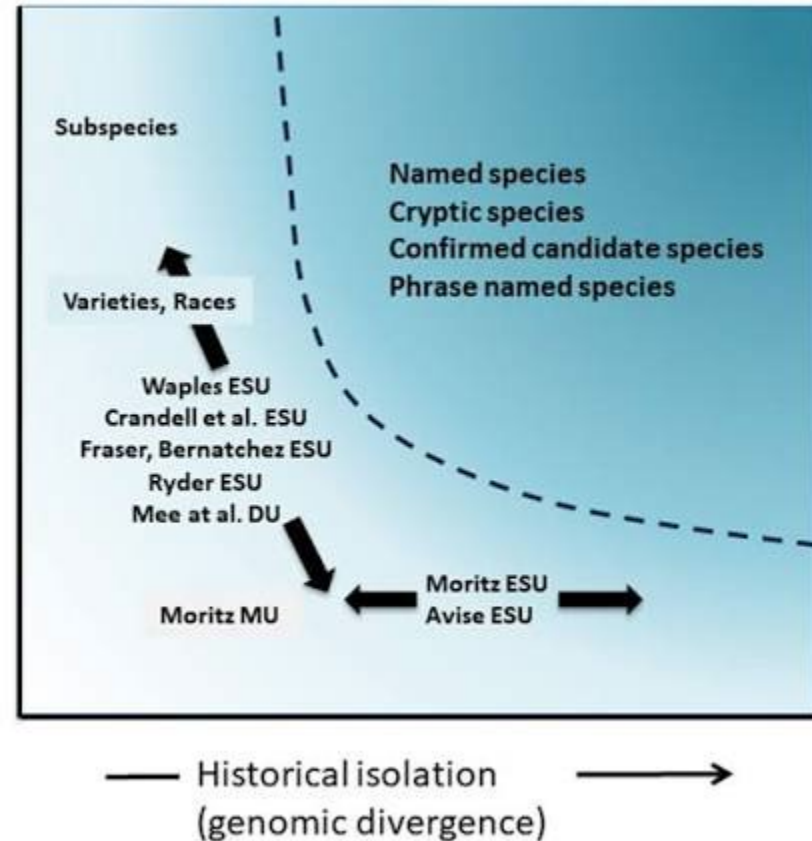
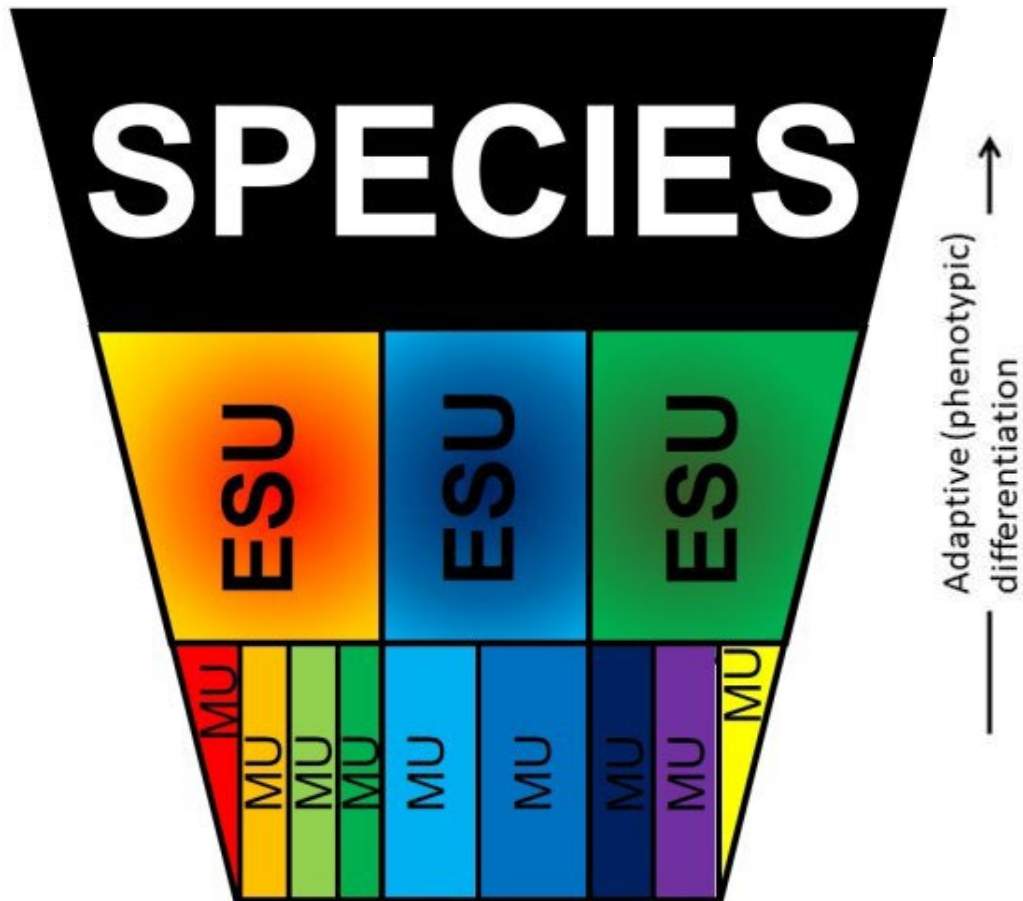


$N=100, m=0.00$



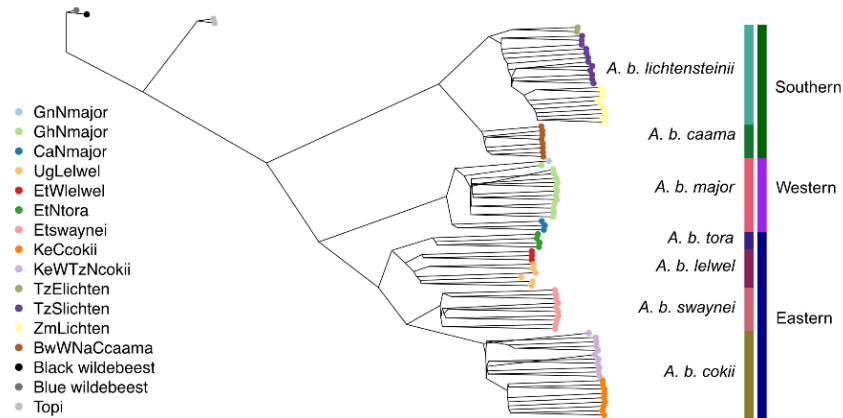
Simulations made with: <https://cjbattey.shinyapps.io/driftR/>



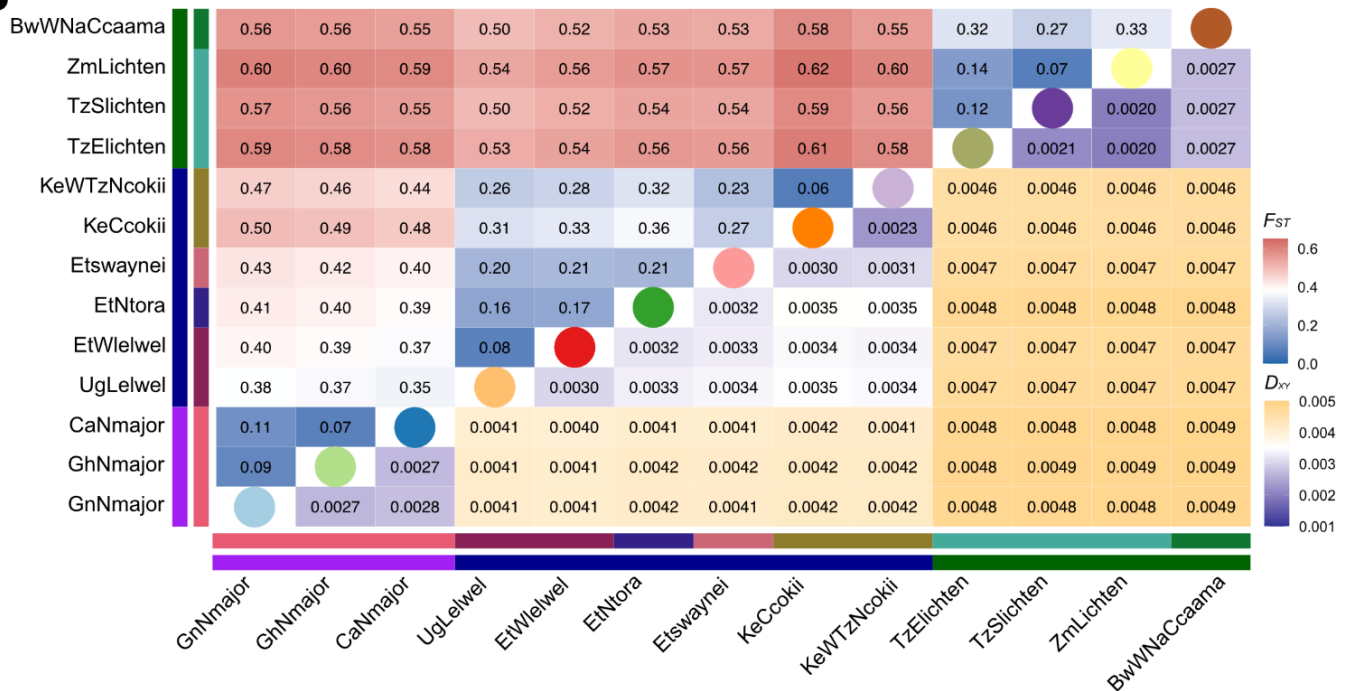


Defining units of biodiversity is an important goal in Conservation Genetics

A

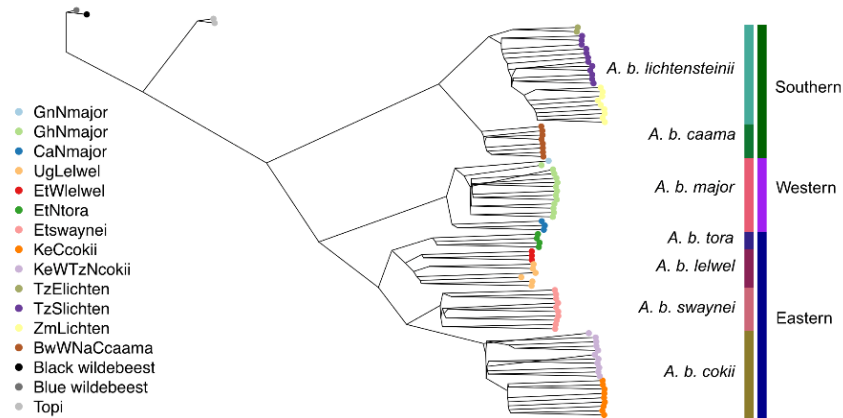


B

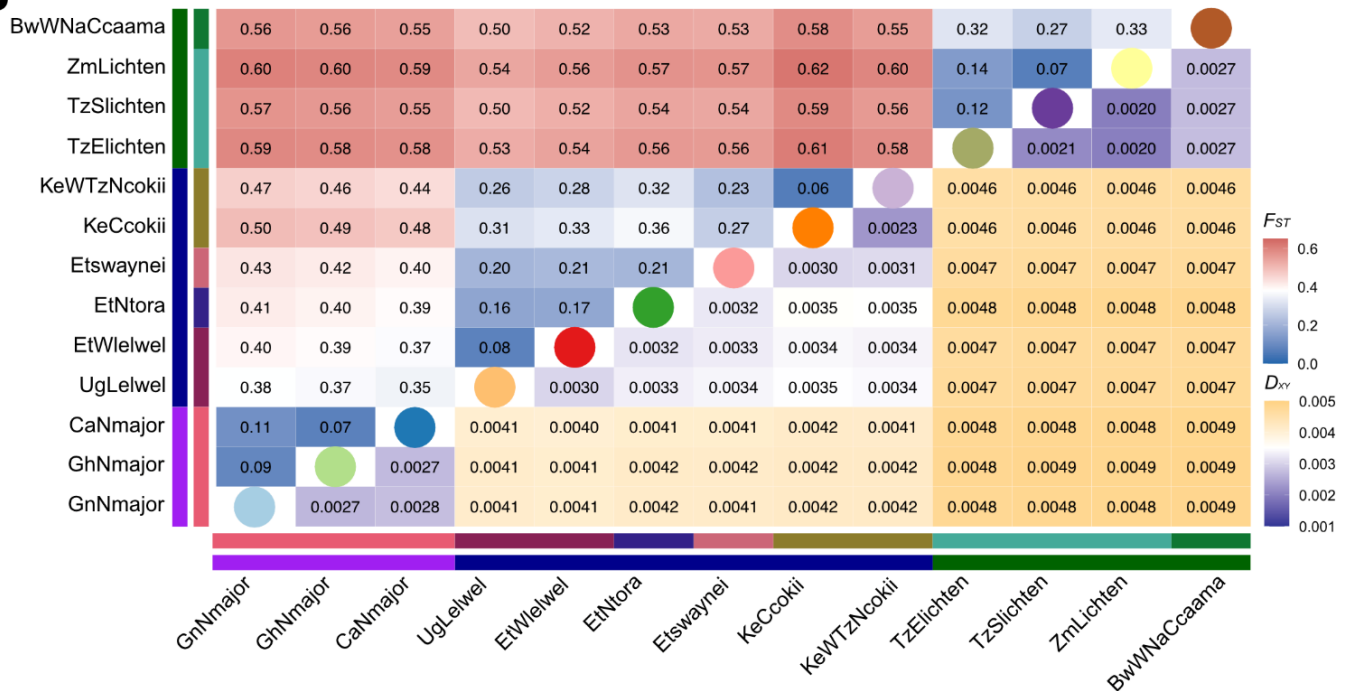


When are populations so genetically differentiated that it is counterproductive to mix them?

A



B



What are the relevant Management Units or Evolutionary Significant Units?
How many species are there?

Recap

- Biodiversity – including genetic diversity – is in decline everywhere, also in Africa.
- Small populations lead to low genetic diversity, which causes multiple problems and can lead to extinction.
- Low genetic diversity is sometimes found in species with relatively large populations today. Genetic monitoring is required to reveal at-risk populations.
- Population fragmentation leads to both population declines and reduction in gene flow. Both decrease the “genetic health” of populations.
- We need to know the genetic structure so we can decide how to manage populations/intraspecific “units”.

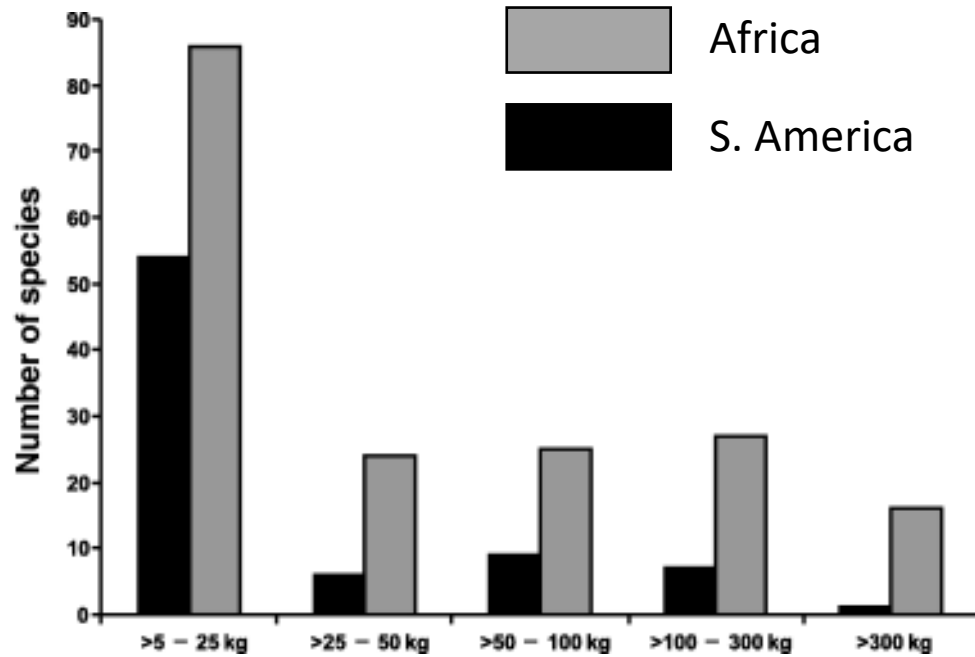


The African Wildlife Genomics project

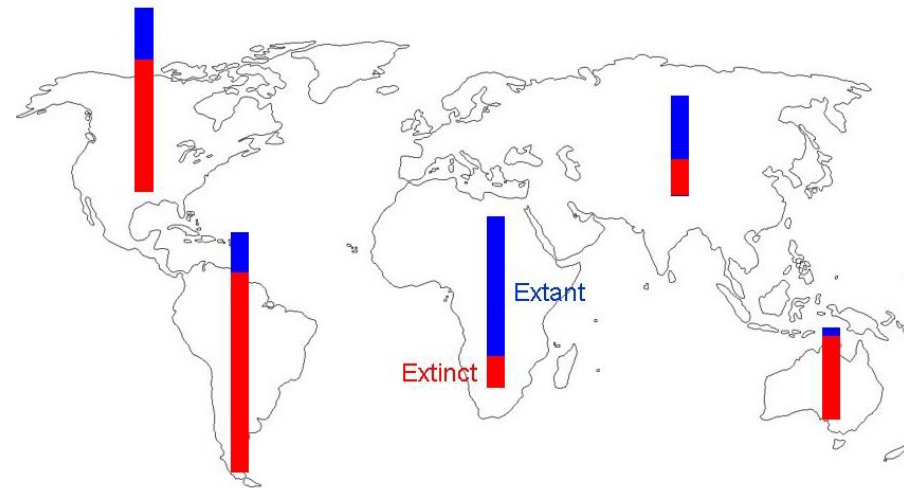
- 40 years of Conservation and Population genetic research @UCPH

Why Africa?





Mammal biodiversity



Why Africa?



"the only [...] remaining portion of the world's pre-human climax community at its tropical richest."

Huxley, 1965



Our last chance to understand and protect?



Our last chance to **understand** and **protect**?



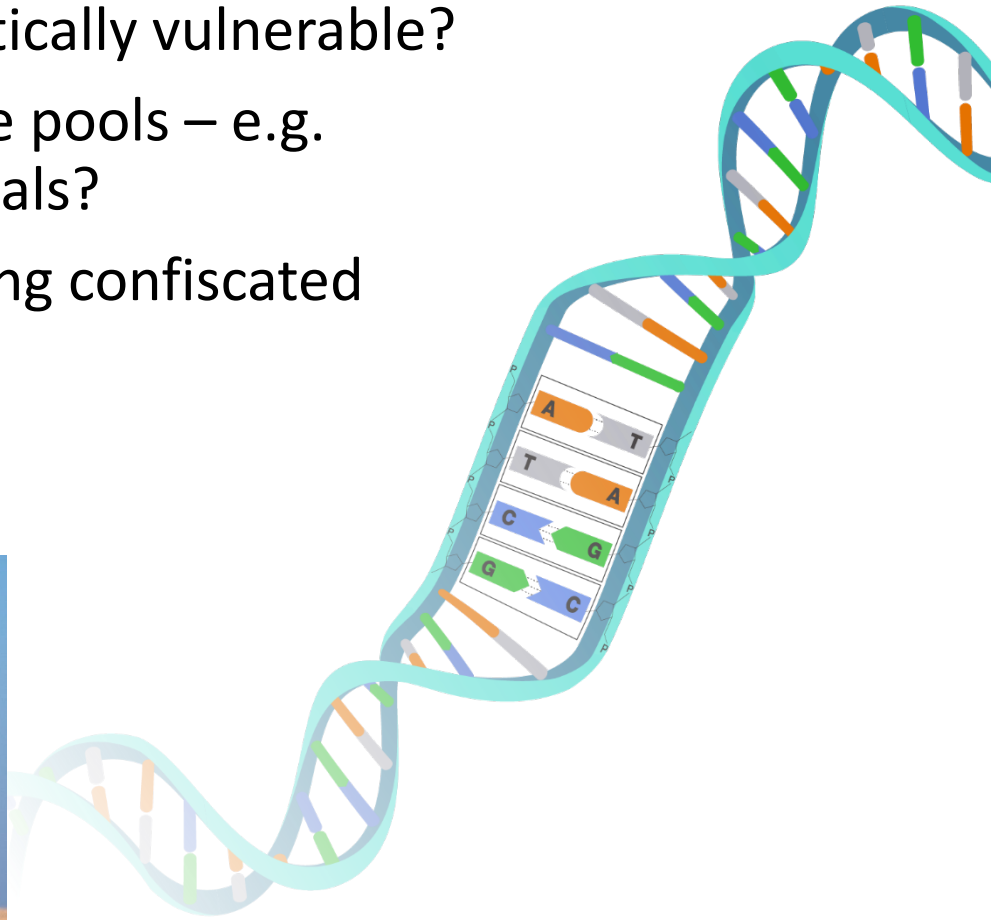
Understanding African wildlife populations

- Where do populations move over time?
- What drives their population sizes and population structure?
- How do populations become differentiated and evolve into new species?
- What are the unique adaptations, and how did these evolve?



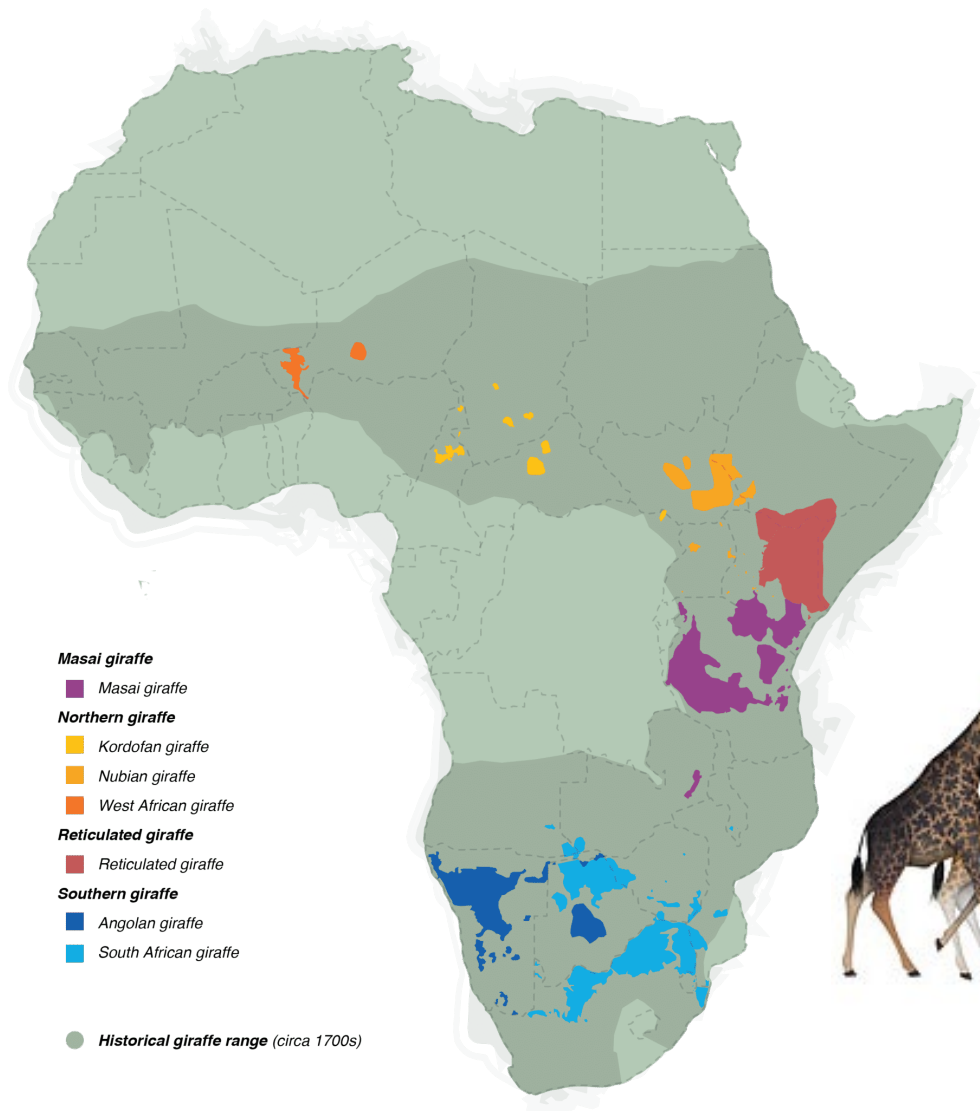
Protecting African wildlife populations

- Which populations are genetically vulnerable?
- How do we manage the gene pools – e.g. where can we move individuals?
- Wildlife forensics: geo-locating confiscated material.



We need many individuals

And, increasingly,
whole genomes.



The African Wildlife Genetics project (since 1994)



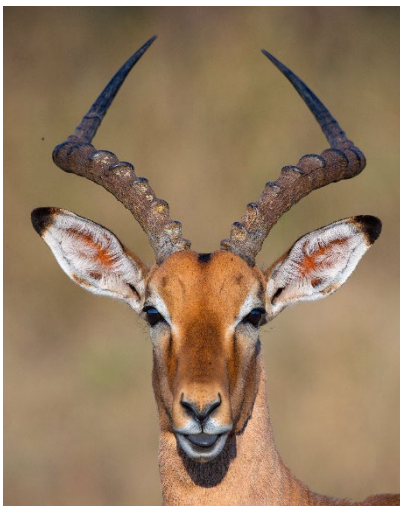
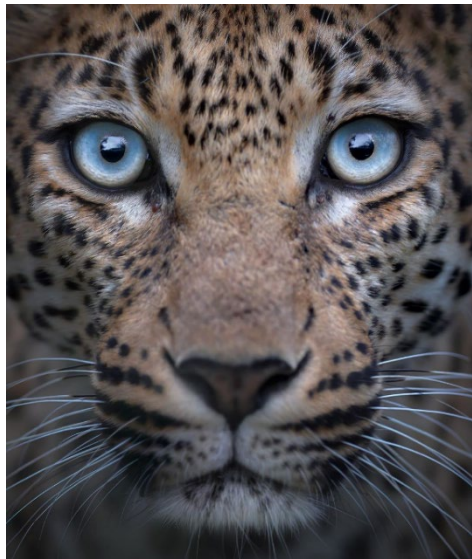
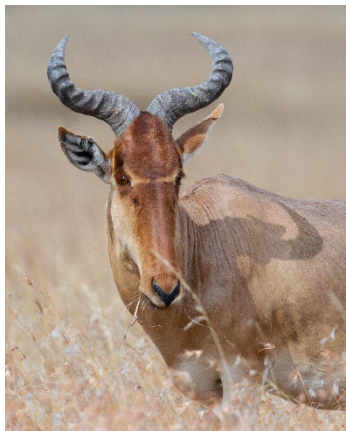
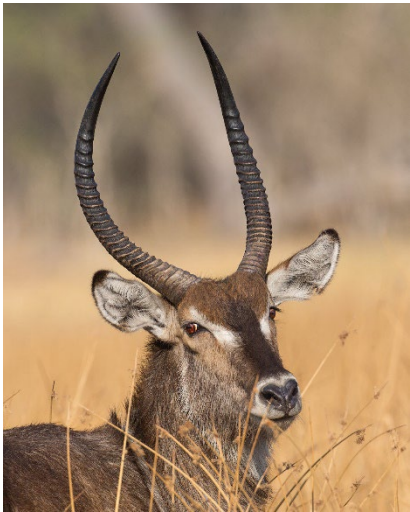
Building biobanks



Computers & servers



Training African scientists



RUMINANT GENOMICS

Science



Large-scale ruminant genome sequencing provides insights into their evolution and distinct traits

MOLECULAR
BIOLOGY AND
EVOLUTION

Warthog Genomes Resolve an Evolutionary Conundrum and Reveal Introgression of Disease Resistance Genes

ORIGINAL ARTICLE

Vicariance followed by secondary gene flow in a young gazelle species complex

Current Biology

Giraffe lineages are shaped by major ancient admixture events

nature communications



Article

<https://doi.org/10.1038/s41467-024-47015-y>

Introgression and disruption of migration routes have shaped the genetic integrity of wildebeest populations

ORIGINAL ARTICLE

MOLECULAR ECOLOGY WILEY

Colonialism in South Africa leaves a lasting legacy of reduced genetic diversity in Cape buffalo

MOLECULAR ECOLOGY WILEY

nature communications



Article

<https://doi.org/10.1038/s41467-023-44105-1>

African bushpigs exhibit porous species boundaries and appeared in Madagascar concurrently with human arrival

Report

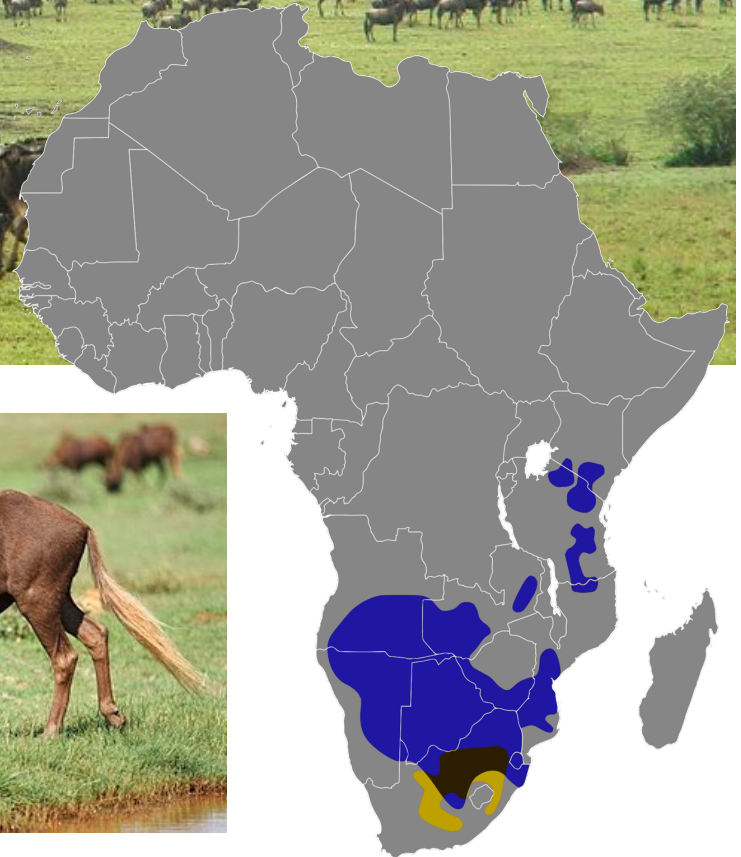
Our vision

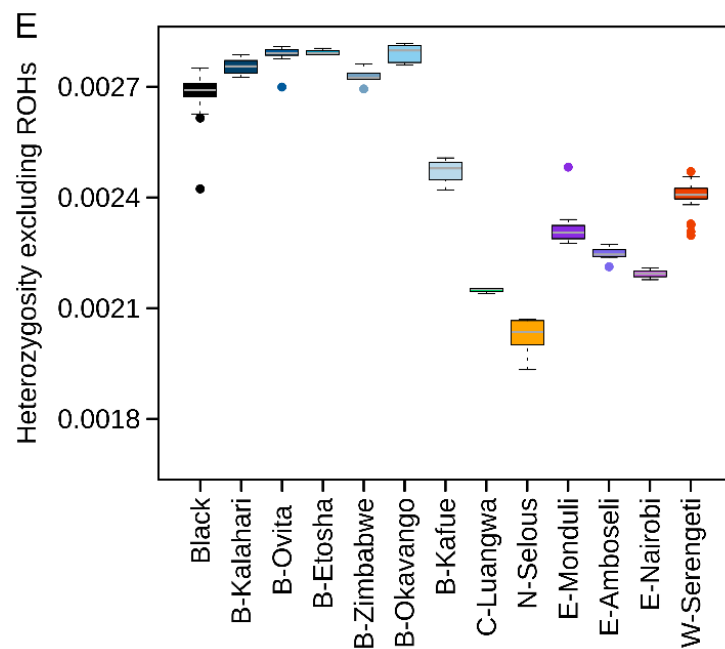
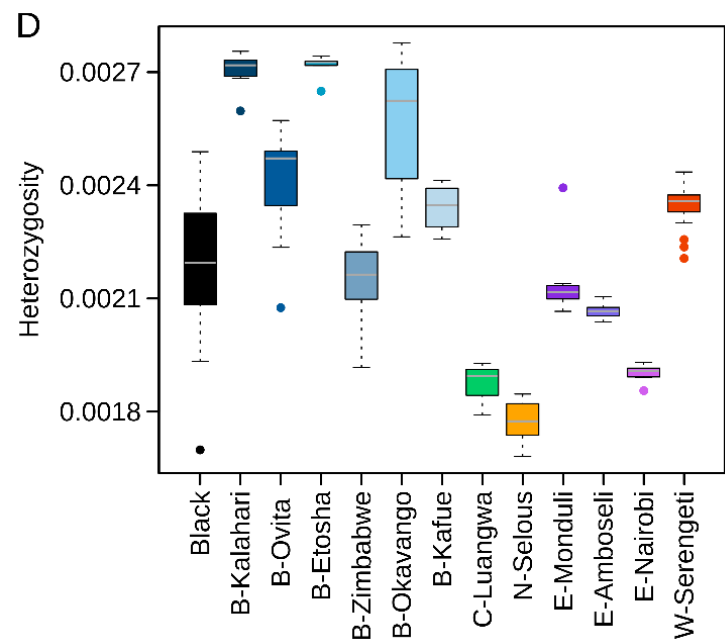
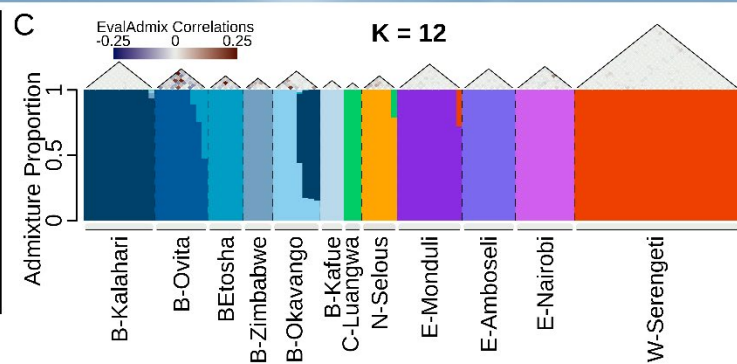
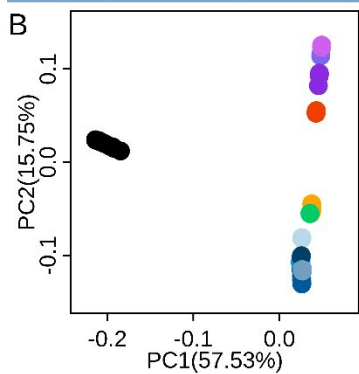
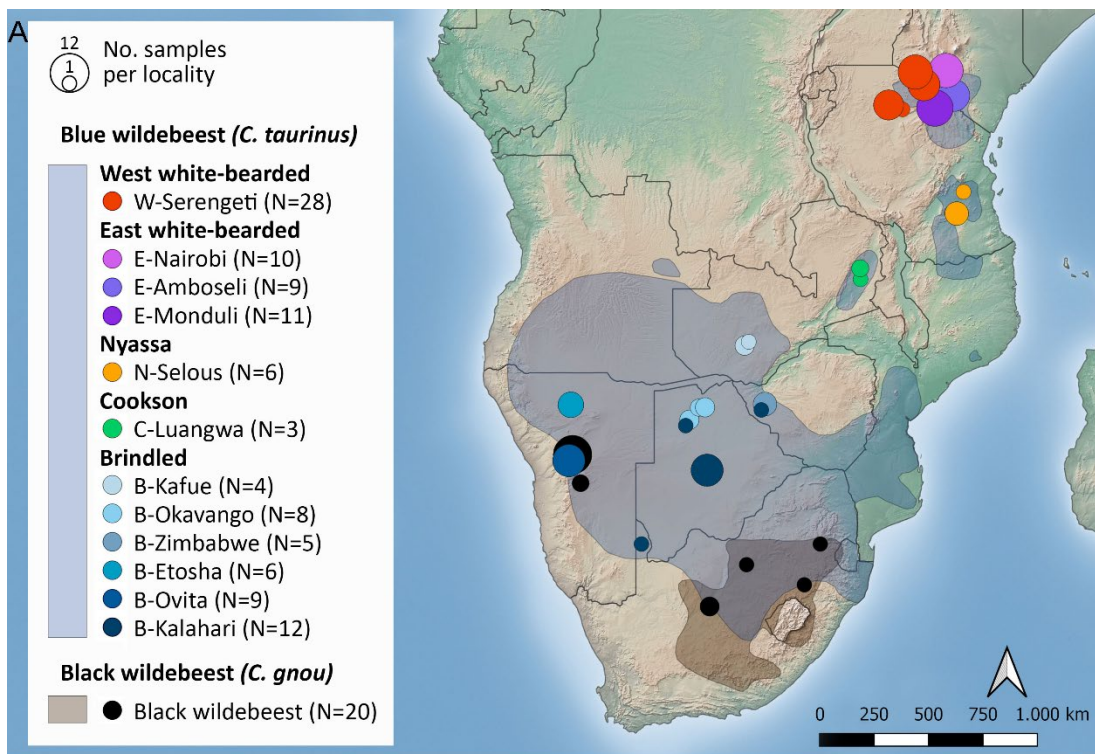
- Sequence 1000s of individuals to at least 15x whole-genome depth.
- Cover 40+ species of large African mammals.
- Develop and optimize best-practices of bioinformatics and population genomics.
- Use this as a roadmap for strategic research and conservation genomics of African mammals.
- Build capacity and partnerships in Africa.



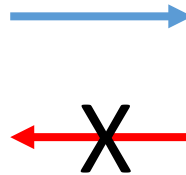
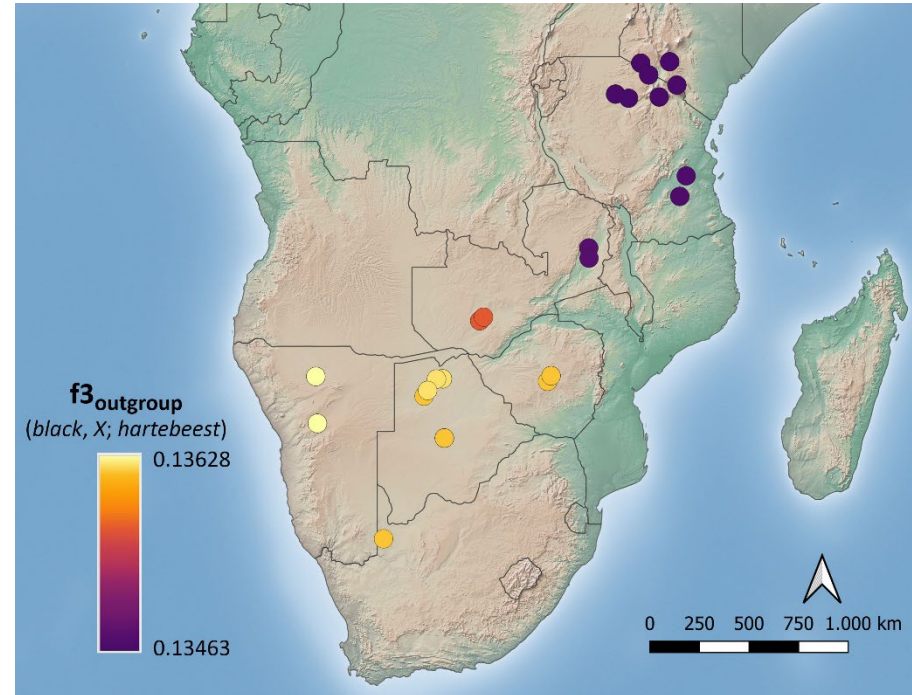
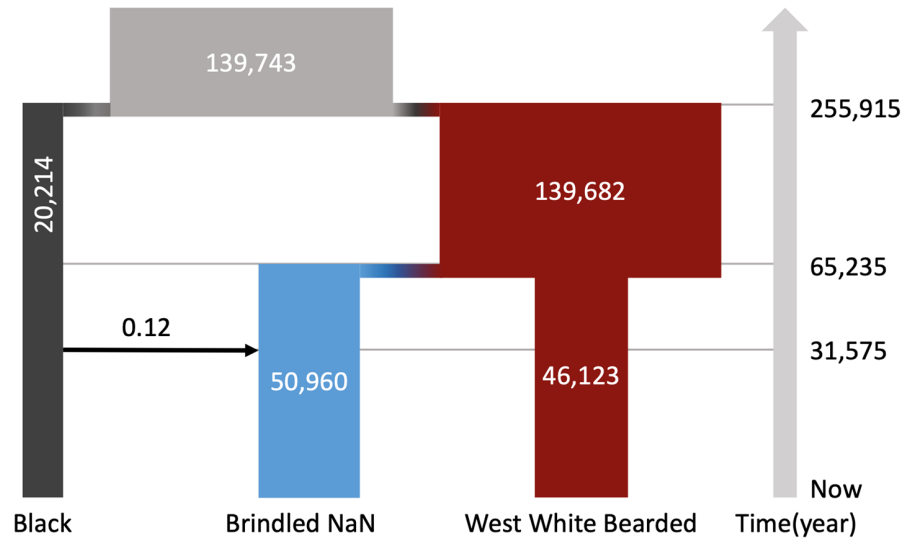
Workflow

Case: Wildebeests

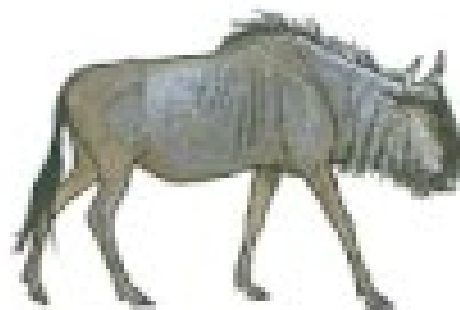
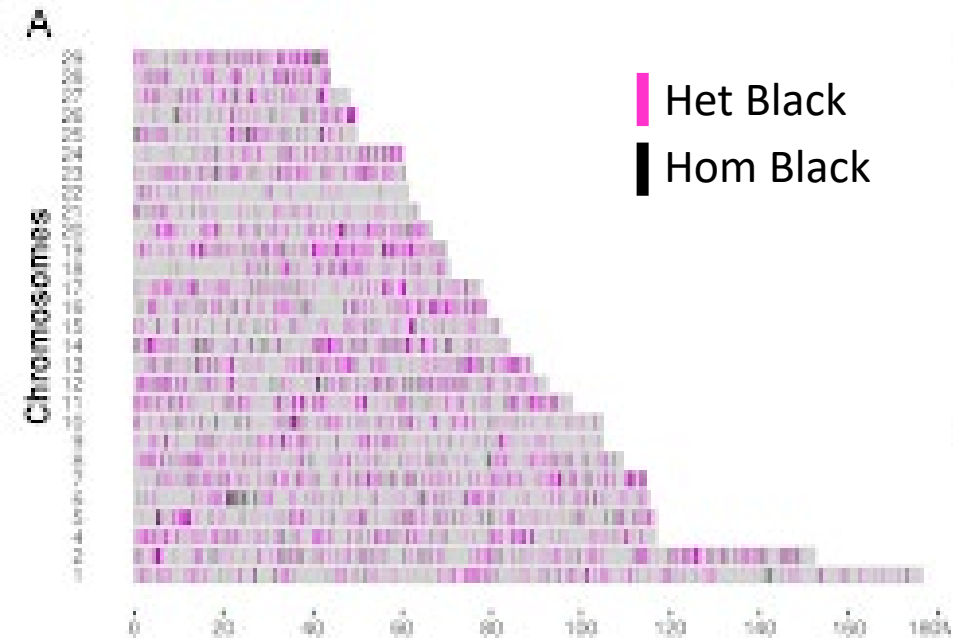




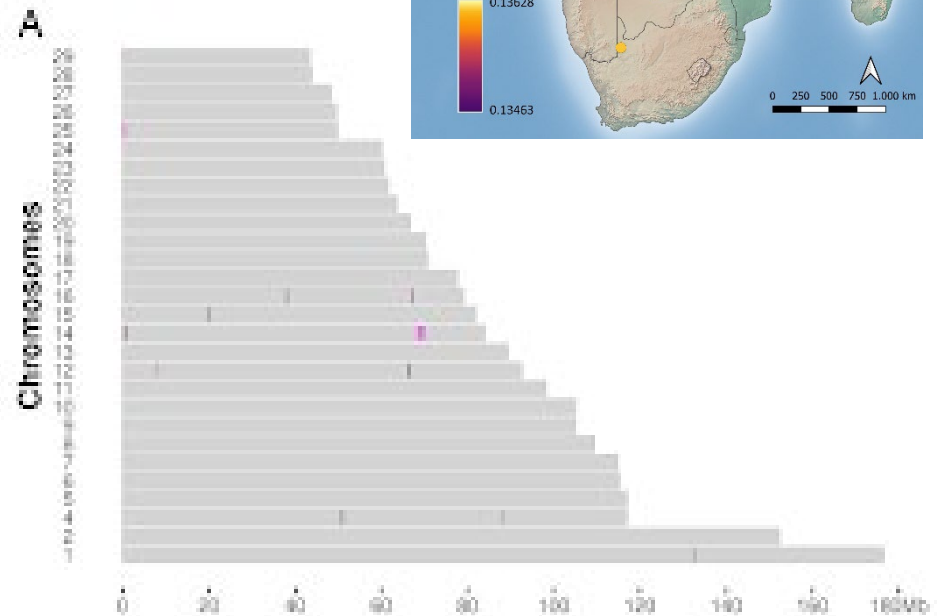
Case: Wildebeests



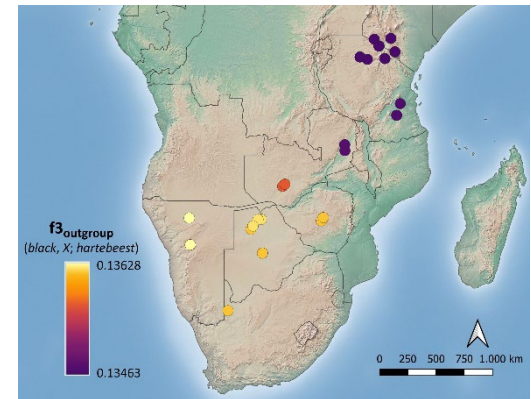
Case: Wildebeests



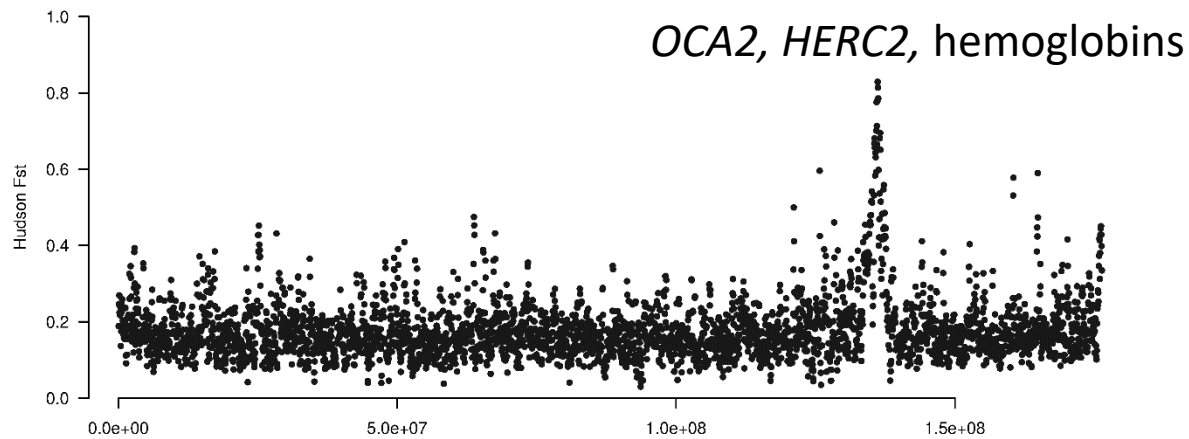
Southern Blue wildebeest



Eastern Blue wildebeest

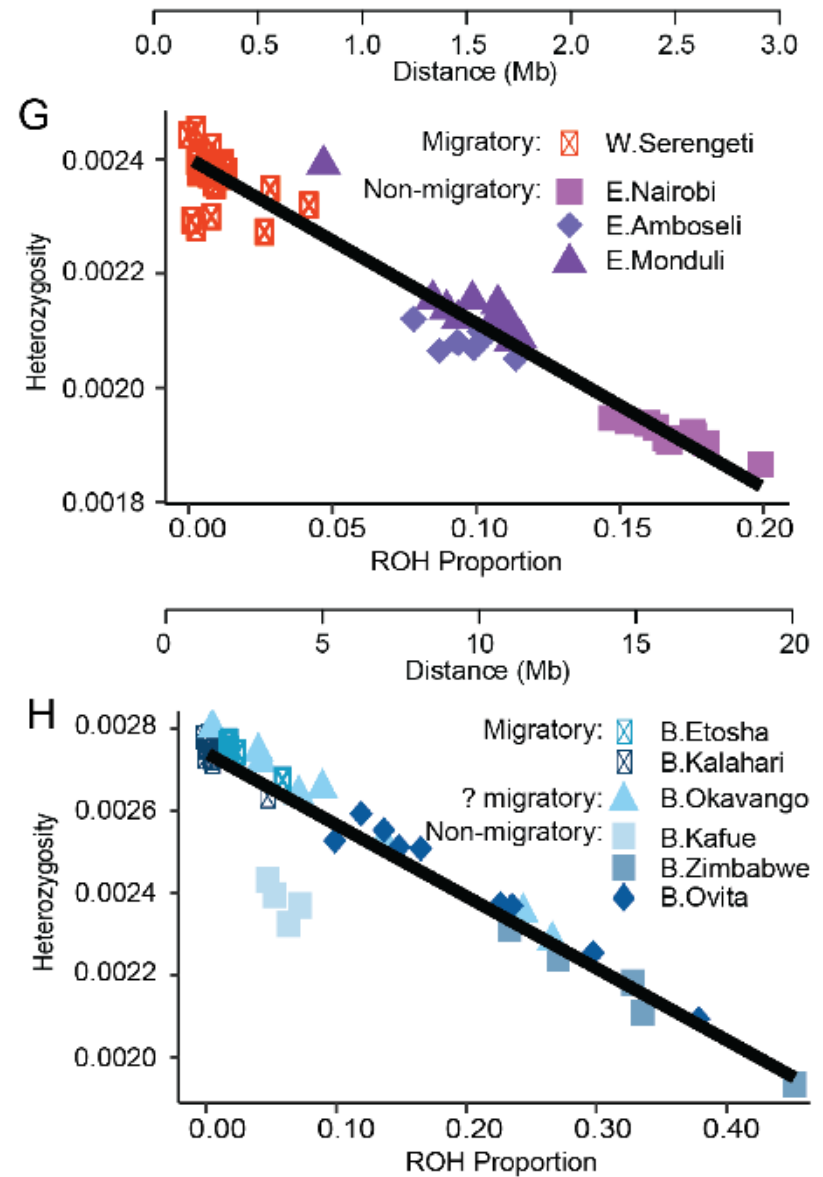
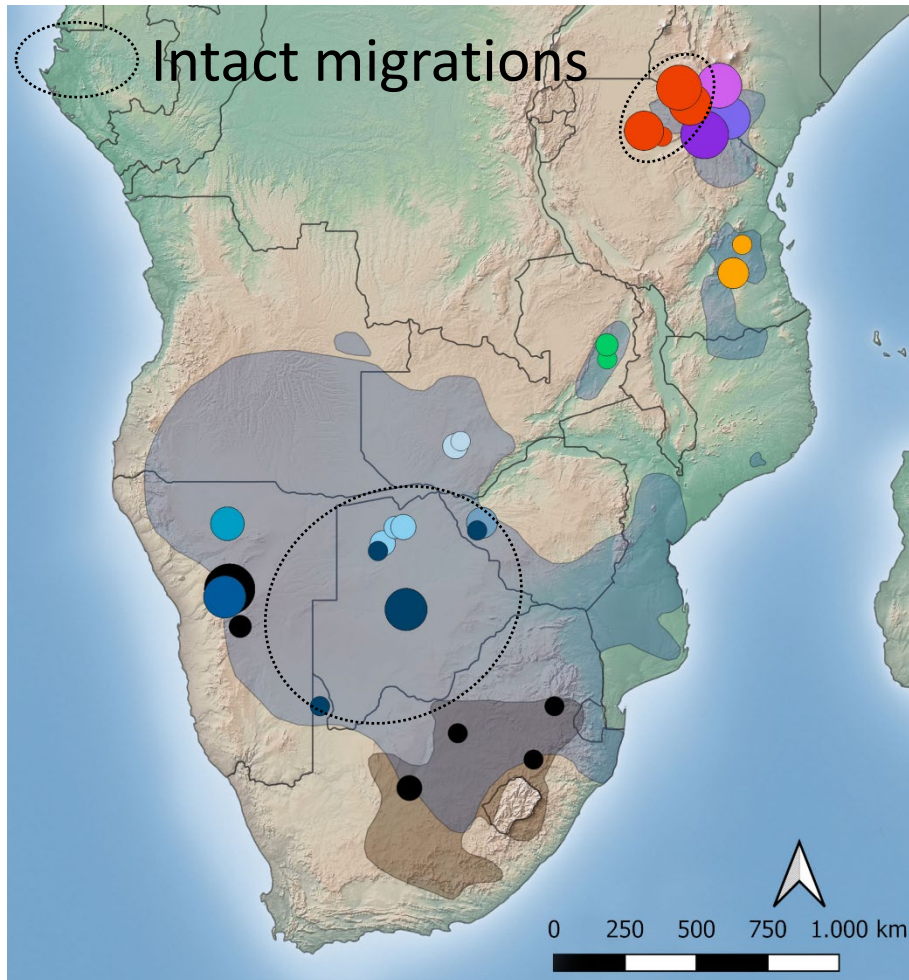


Case: Wildebeests

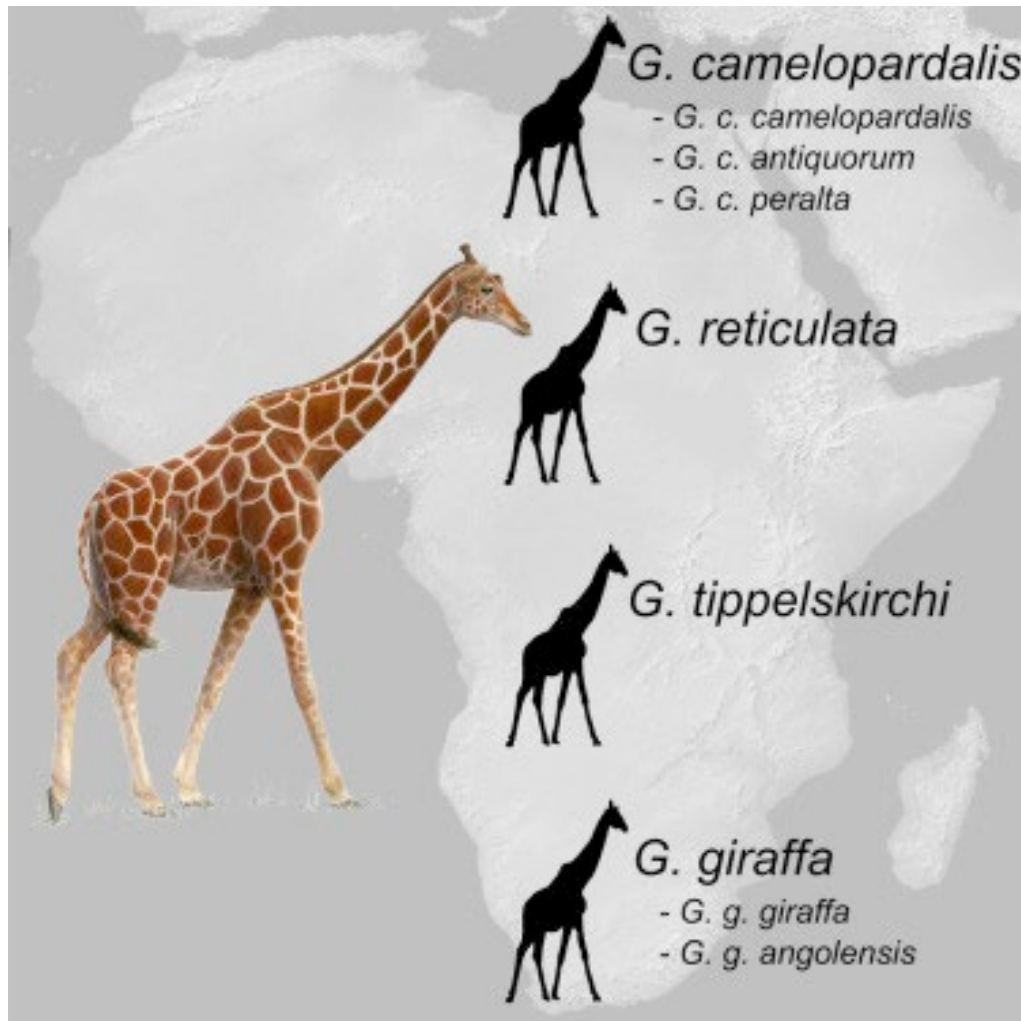


Peak of differentiation in color coding and aerobic genes

Case: Wildebeests



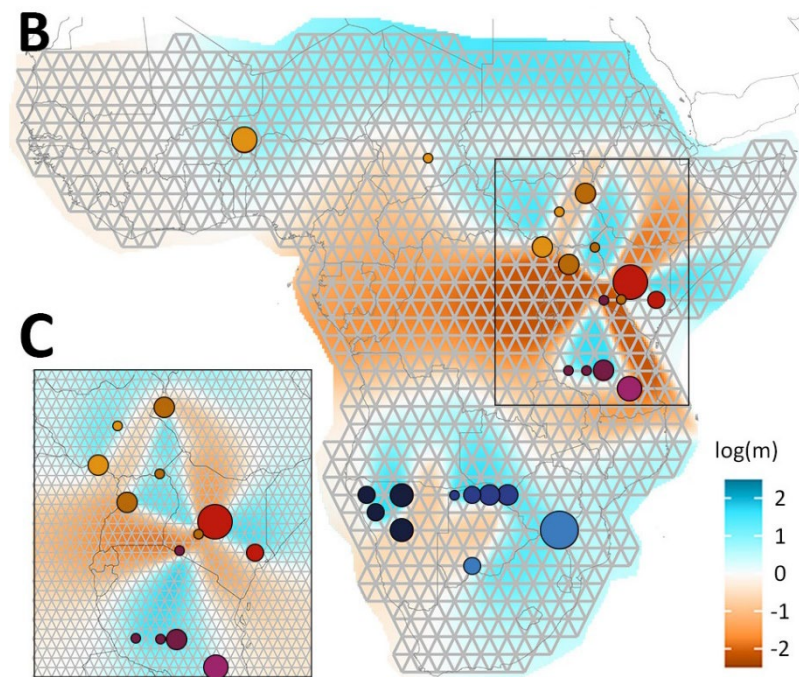
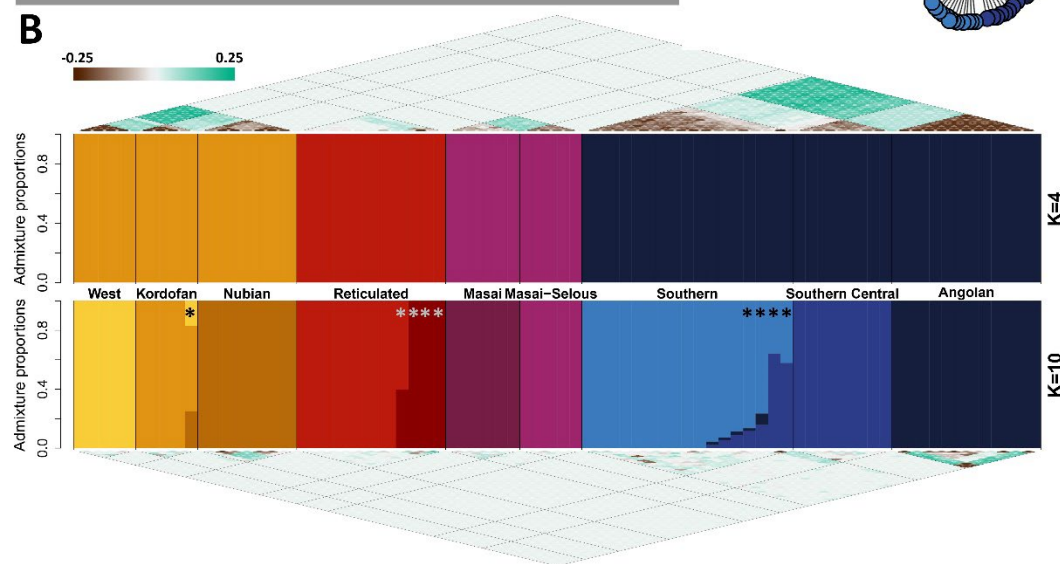
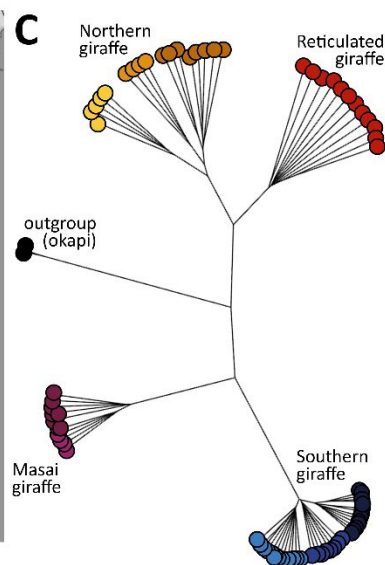
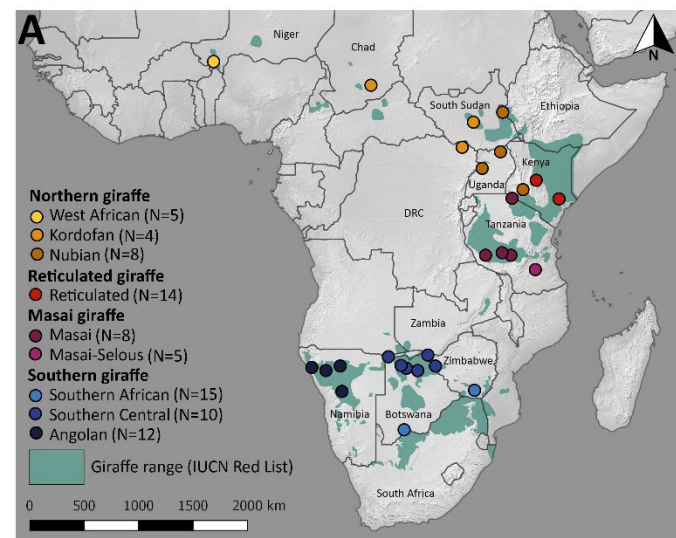
Case: Giraffes



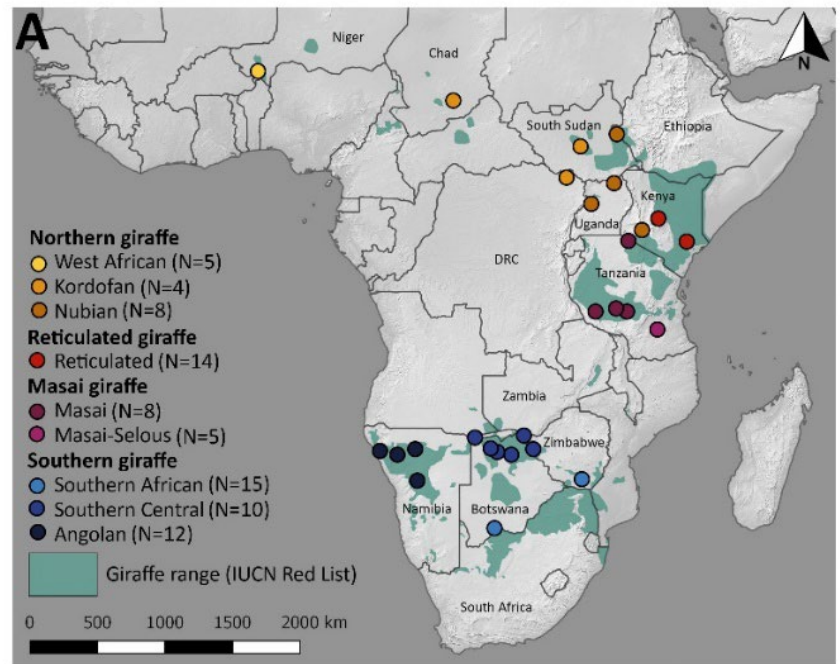
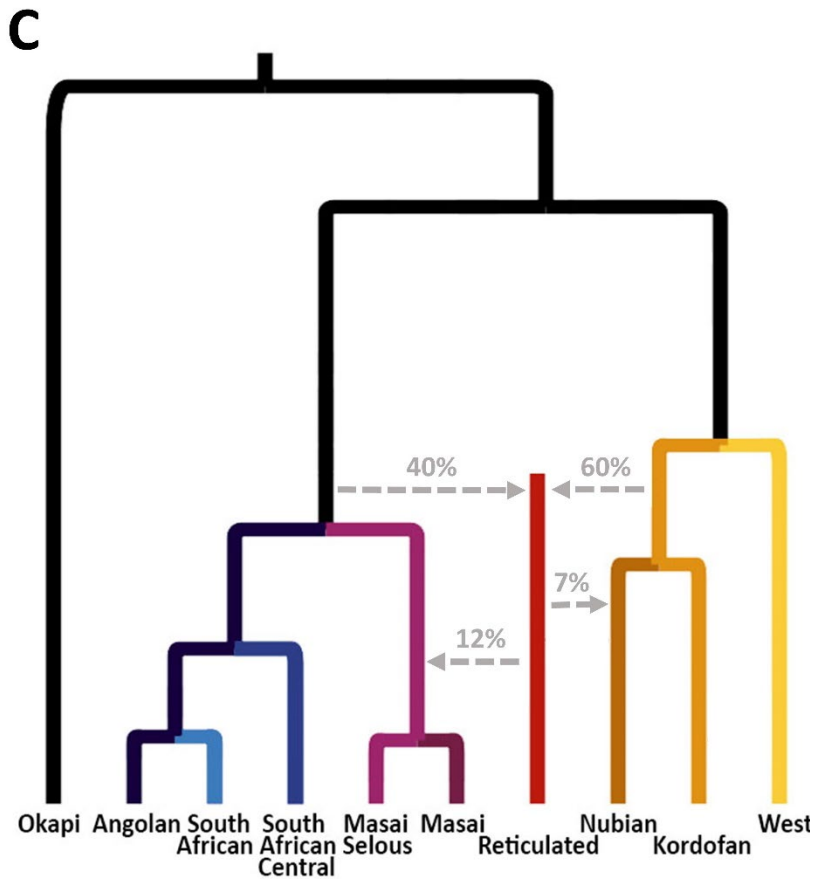
One species –
or many??



Case: Giraffes

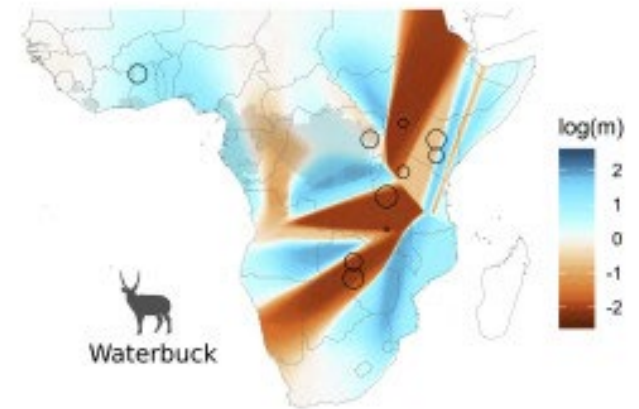
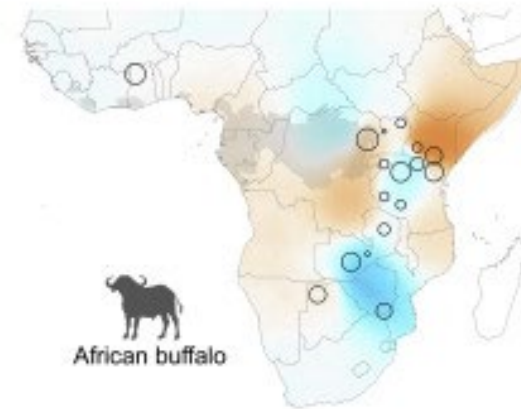
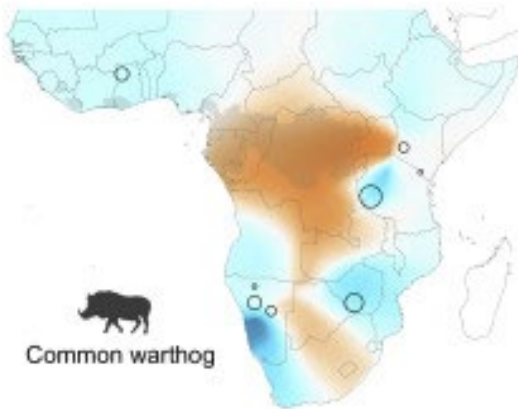
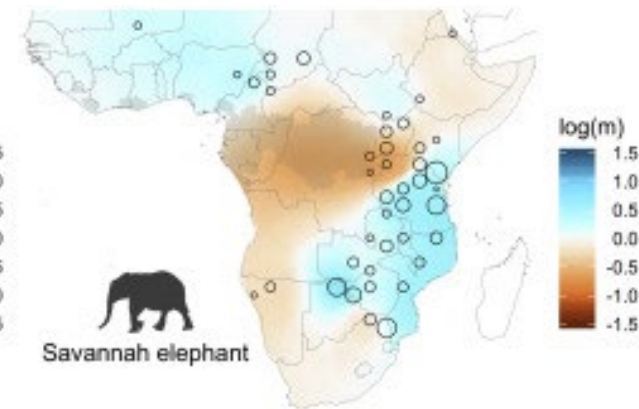
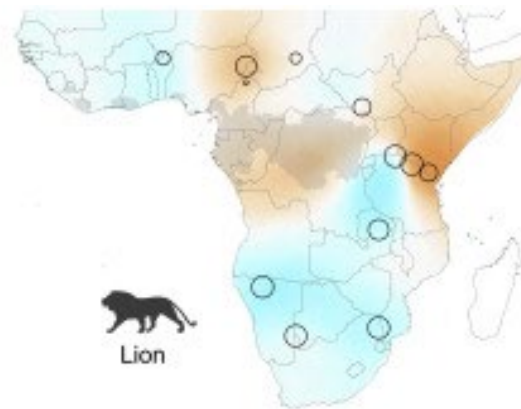
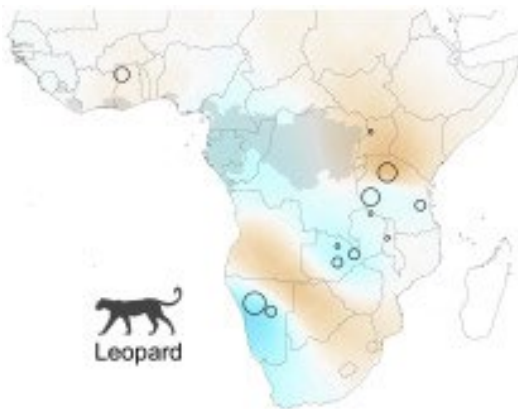


Case: Giraffes



Prolific gene flow – Reticulated is a hybrid

Cross-species comparisons



equatorial rainforest

We work with African partners

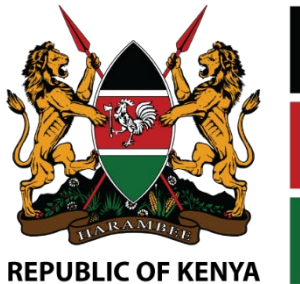
- Build collaborations for further funding, sample collection and biobanking.
- Making it easier for African researcher to whole-genome sequence.
- Identify focal conservation problems/species/populations to tackle the most important problems first.
- Translate the science into action.
- Train the next generation of African scientists.

African BioGenome Project

Genomics for the future of biological diversity across Africa



Building African collaborations



**WILDLIFE
RESEARCH
& TRAINING
INSTITUTE**

Discover Beyond



Acknowledgments



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